

Lesson Four

Plan Software Project

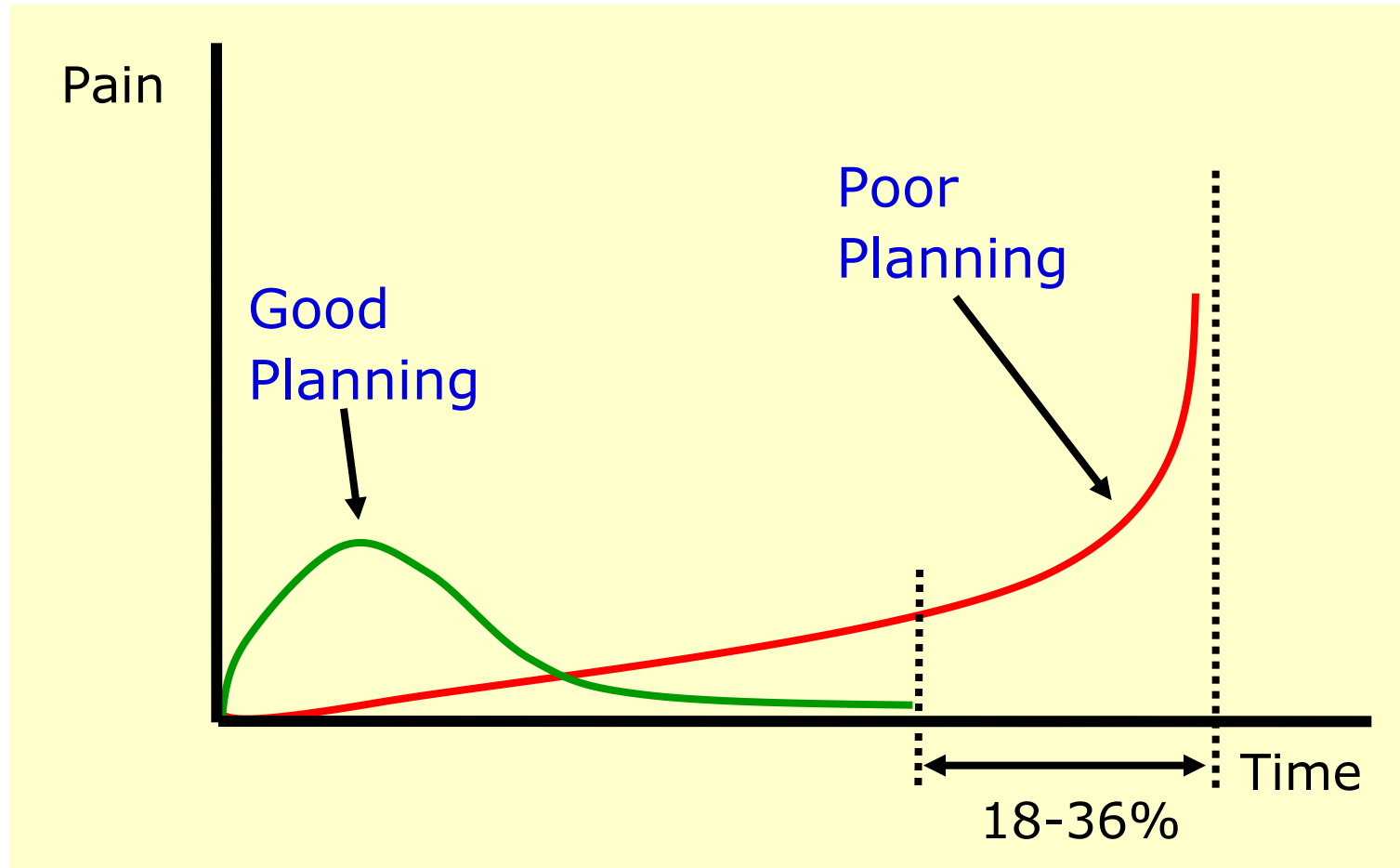
Effective Project Management: Traditional, Agile, Extreme

Ch07: How to Plan a TPM Project

***Presented by
Bowen DU***

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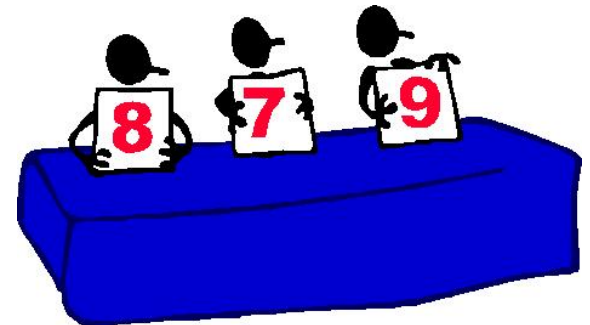
Pain Curve



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The Importance of Planning

- Planning Reduces Uncertainty
- Planning Increases Understanding
- Planning Improves Efficiency



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Traditional Project Planning Tools

- Sticky notes (assorted colors)
- Marking pens (assorted colors)
- Plenty of whiteboard space



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Case: uses for Sticky Notes

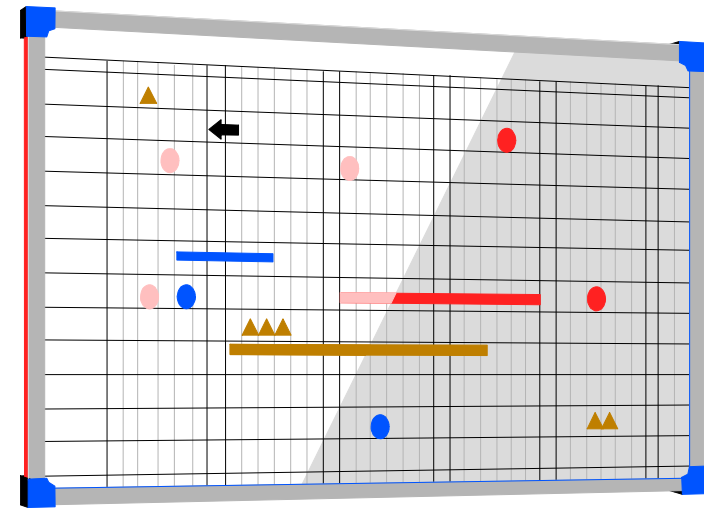
- Task ID
- Unique task name
- Task duration
- Task labor
- Resource requirements
- Task manager
- ES, EF, LS, and LF (calculated values)
- Critical Path (calculated)



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Case: uses for the Whiteboard

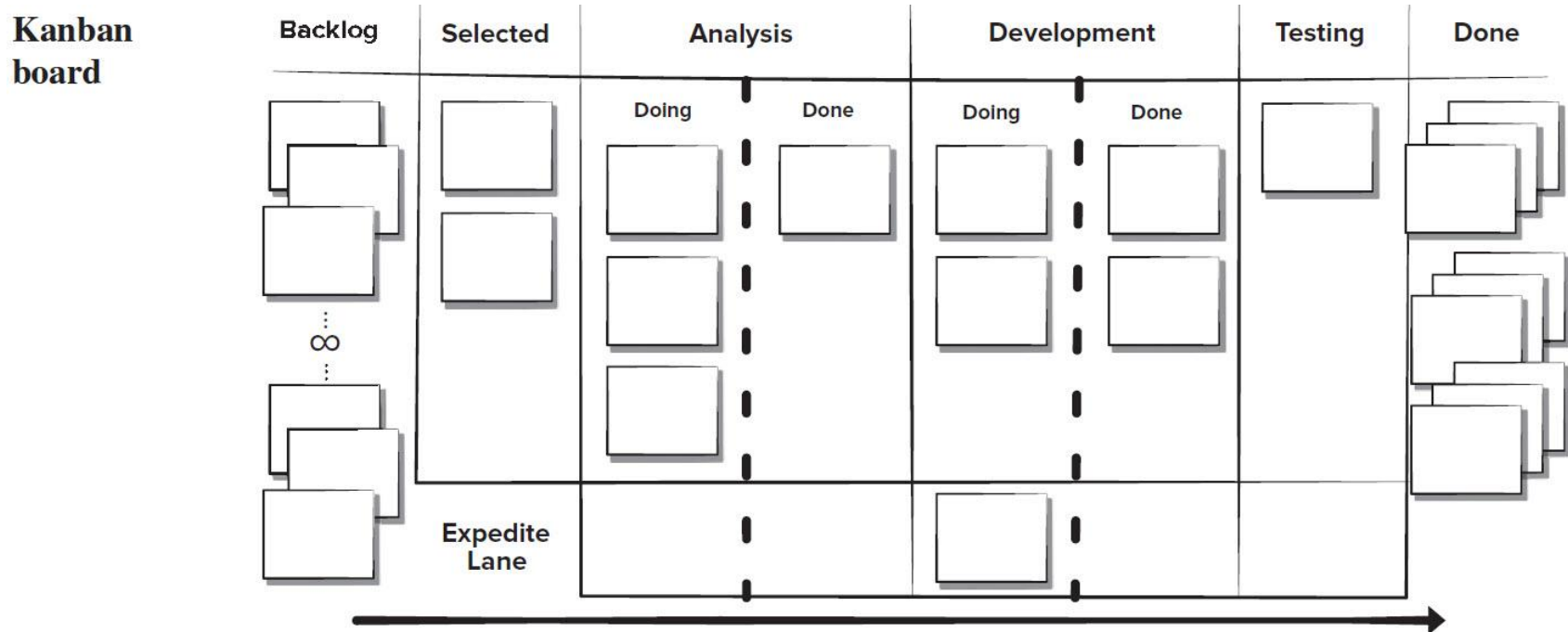
- Project Overview Statement
- Work Breakdown Structure
- Dependency Diagram
- Initial Project Schedule
- Final Project Schedule
- Resource Schedule
- Issues Log
- Updated Project Schedule



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Case: Kanban

- An agile method of software process



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How much Time Should Planning Take?

- Very small projects < ½ day
- Small projects < 1 day
- Medium projects 2 days
- Large projects 3-4 days
- Very large projects ???



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Form: The JPPS Meeting Components

- Attendees
- Facilities
- Equipment
- Agenda
- Deliverables
- Project Proposal

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Form: The JPPS Attendees

- Facilitator
- Project manager
- Another project manager
- JPP consultant
- Technographer
- Core project team
- Client representative
- Resource managers
- Project champion
- Functional managers
- Process owner

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Form: The Complete JPPS Agenda

- Session #1
 - Negotiate the COS
 - Build the RBS
- Session #2
 - Write the POS
- Session #3
 - Part 1: Kick-off
 - Part 2: Working session

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Form: Who's Involved in the Planning Process?

- Session #1 Project Manager
Client
- Session #2 Project Manager
Client
Project Team (known members)
- Session #3 Project Manager
Entire Project Team

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Case: The JPPS Agenda – Session #3 Planning Kick-off

- Introduction of Sponsor
- Sponsor overview of the project and its importance to the company/division/departement
- Introduction of the client co-project manager
- Introduction of the IT co-project manager
- Introduction of the core project team
- Introduction of the planning facilitation team (facilitator and technographer)

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Case: JPPS Agenda – Session #3 Working session

- Validation & prioritization of requirements
- Overview of the project planning approach to be taken
- Generation and validation of the WBS
- Estimate duration, labor and resource requirements
- Creation of the dependency diagram
- Discuss critical path and project completion date
- Analysis and schedule compression as needed
- Identification of project risks and mitigation plans
- Planning team consensus on project plan contents
- Adjournment

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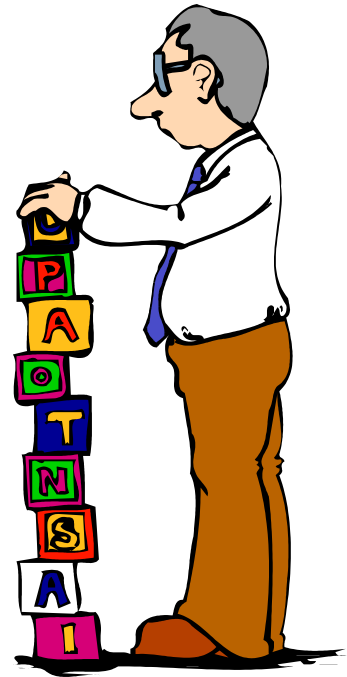
Planning Session Activities

- Review the POS for clarity
- Create Work Breakdown Structure (WBS)
- Estimate task duration and resource needs
- Construct project network diagram
- Determine Critical Path (CP)
- Revise and approve project schedule
- Finalize resource schedule
- Gain consensus on the project plan

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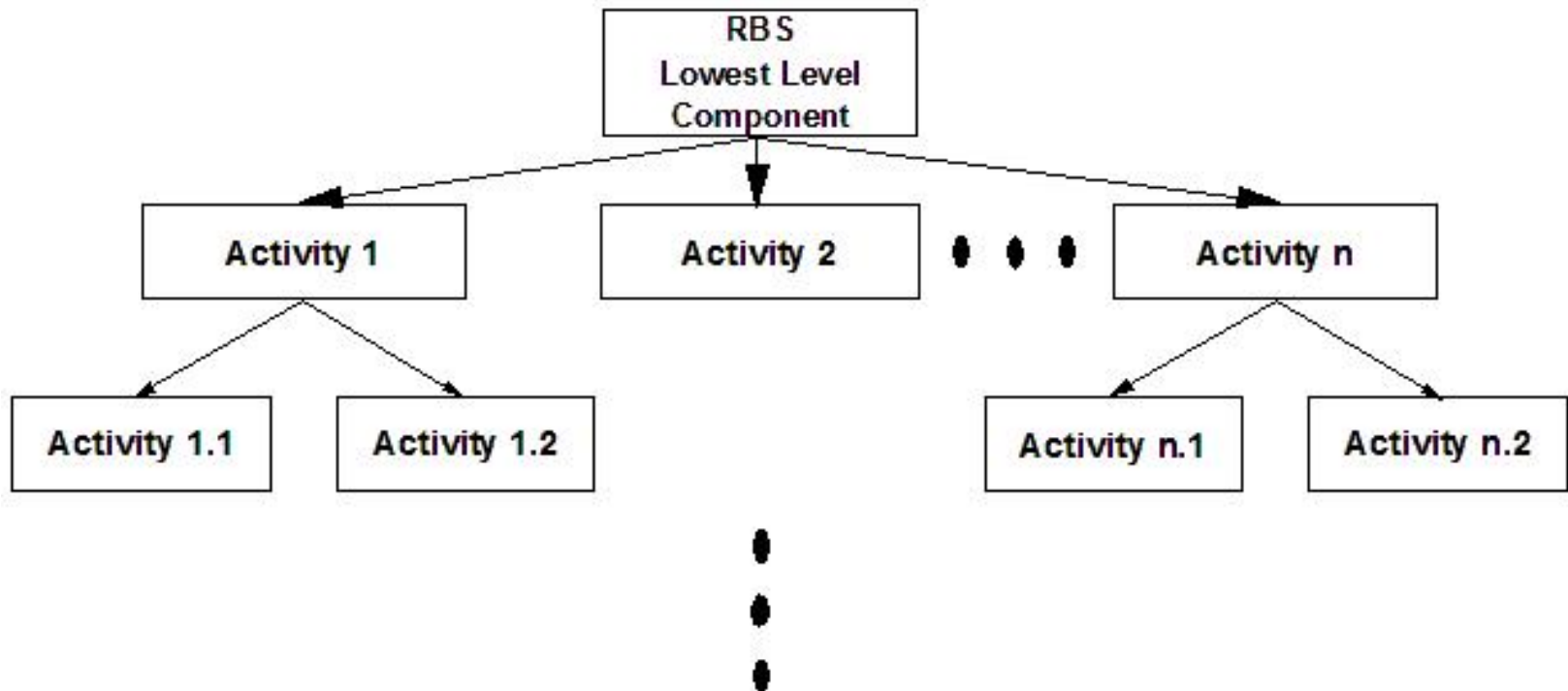
Definition of the WBS

The Work Breakdown Structure (WBS) is a hierarchical description of all of the work that must be done to meet the needs of the client.



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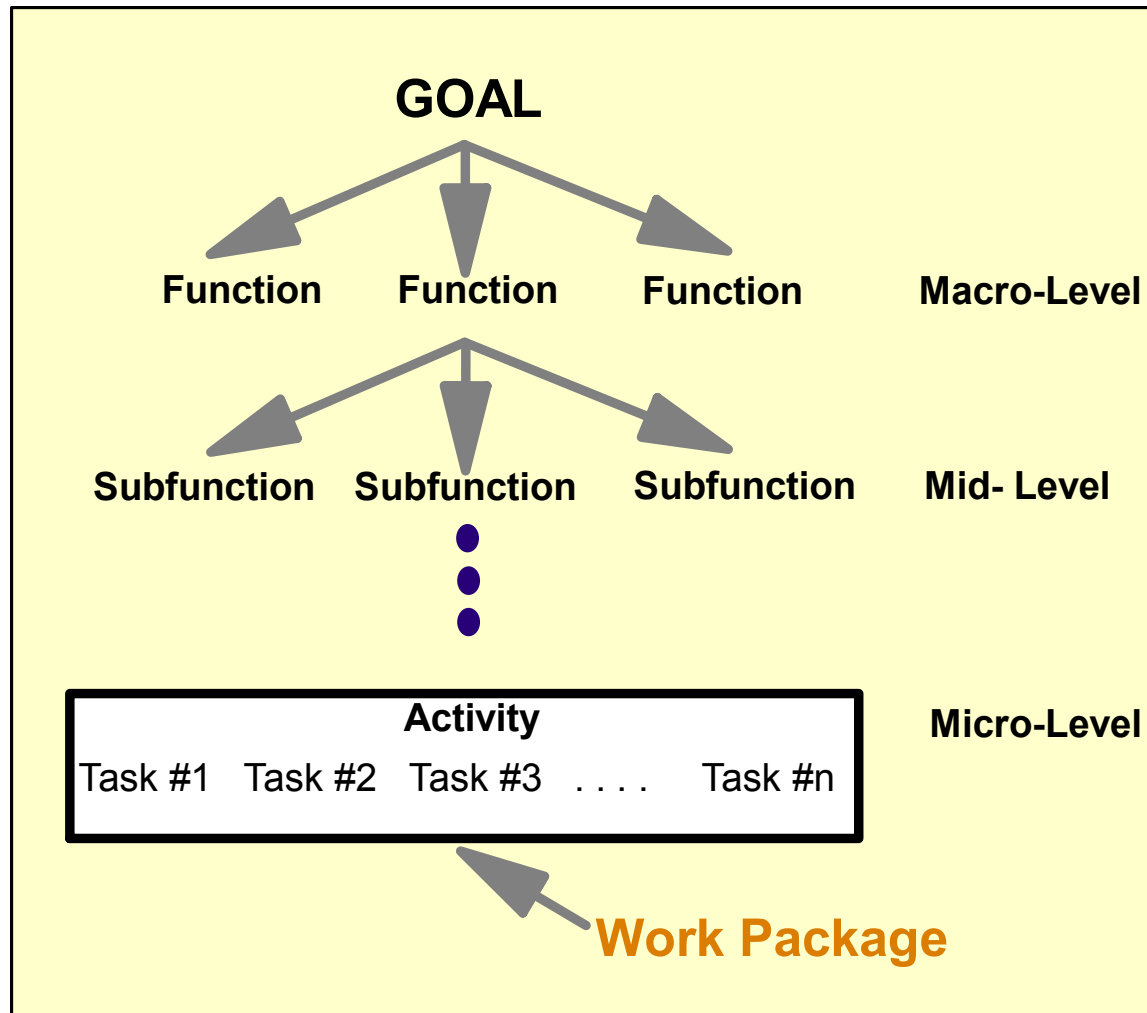
RBS and WBS



Each activity is a further decomposition of the WBS.

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Hierarchical visualization of the WBS



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Uses for the WBS?

- Thought-process tool
- Architectural design tool
- Planning tool
- Project-status-reporting tool

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Generating the WBS

- Converting the RBS to the WBS (Fig7.2 textbook)
 - Team Approach
 - Sub-team Approach
- Top-Down Approach
 - Team Approach
 - Sub-team Approach
- Bottom-Up Approach



Which is best?
When?
Why?

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WBS Completion Criteria

- 1. Status and completion are measurable.
- 2. The activity is bounded.
- 3. The activity has a deliverable.
- 4. Time and cost are easily estimated.
- 5. Activity duration is within acceptable limits.
- 6. Work assignments are independent.

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WBS Completion Criteria – The 7th criterion

- The client didn't participate in building the WBS like you expected.
- You have an uncomfortable feeling.
- Scope change is likely to be a big part of the project.
- Choose a PMLC model that accommodates frequent change.

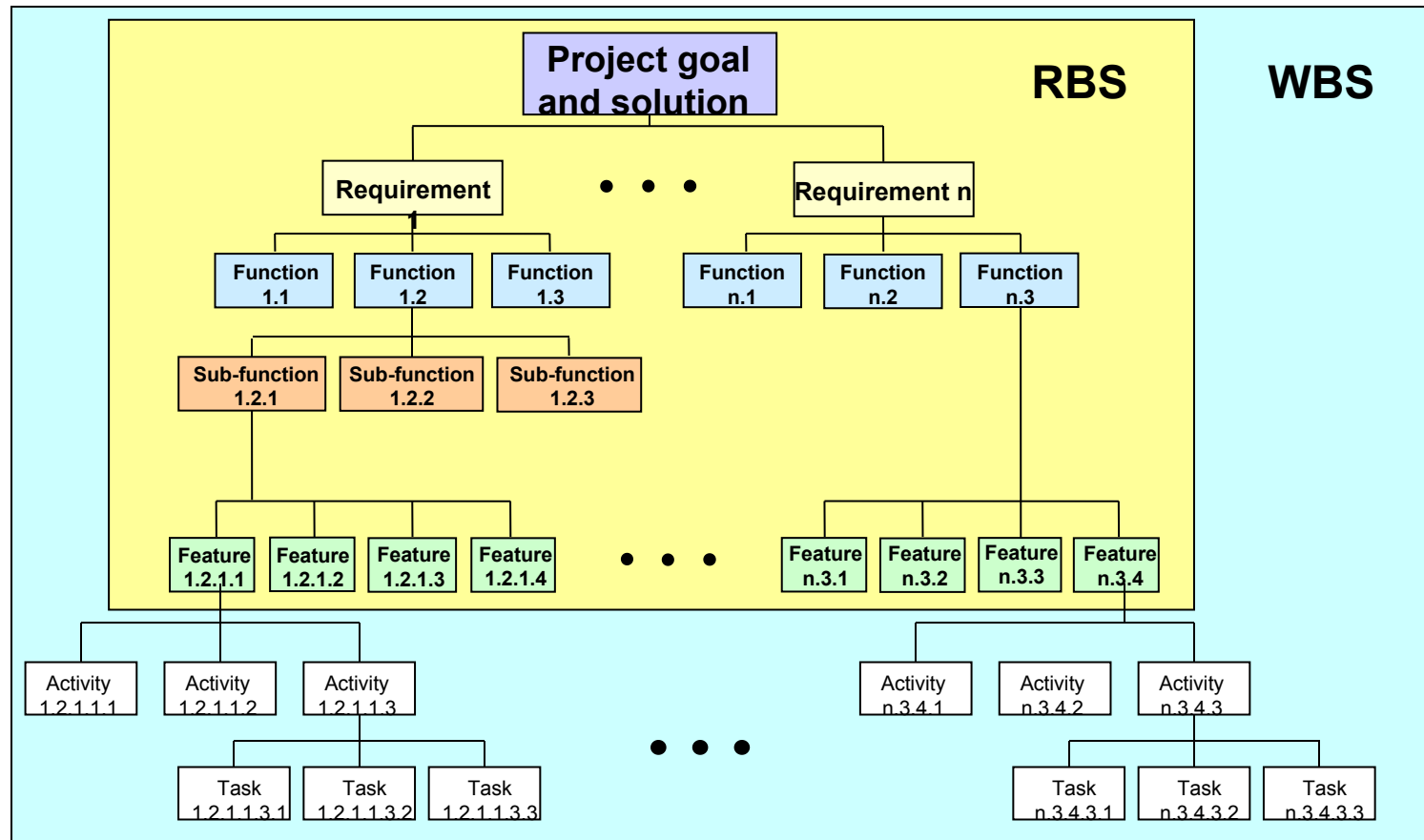
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WBS Completion Criteria – Exceptions

- Stopping Before Completion Criteria Are Met.
- Decomposing Beyond(迟于) Completion of the Criteria
 - Short duration projects
 - High risk activities
 - Large duration variance

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RBS is a subset of the WBS



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Approaches to Building the WBS

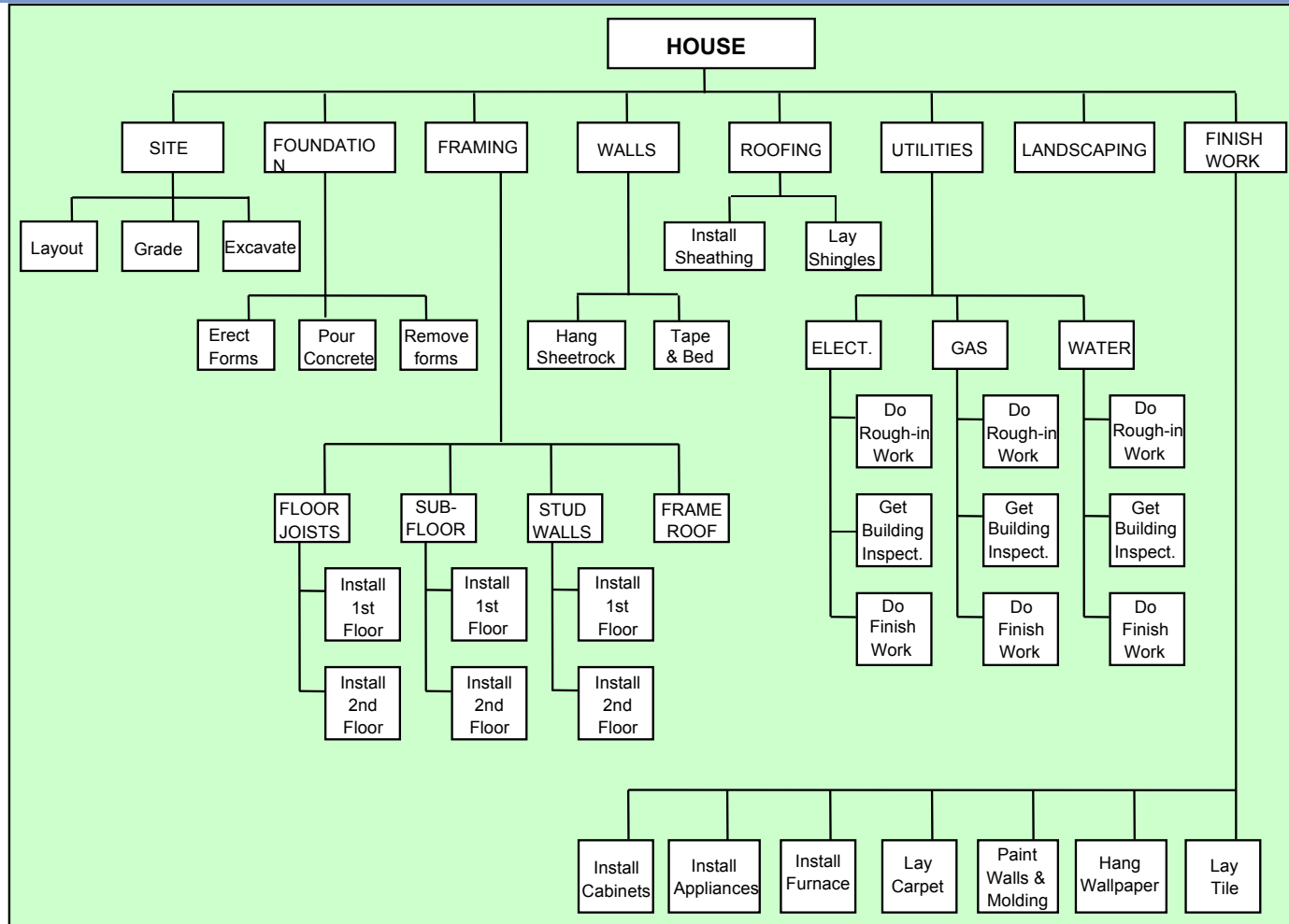
- Noun-type Approaches
 - Physical components
 - Functional components
- Verb-type Approaches
 - Design-build-test-implement
 - Objectives
- Organizational approaches
 - Geographic
 - Departmental
 - Business process



**Which is best?
When?
Why?**

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Case: Graphical WBS for a House



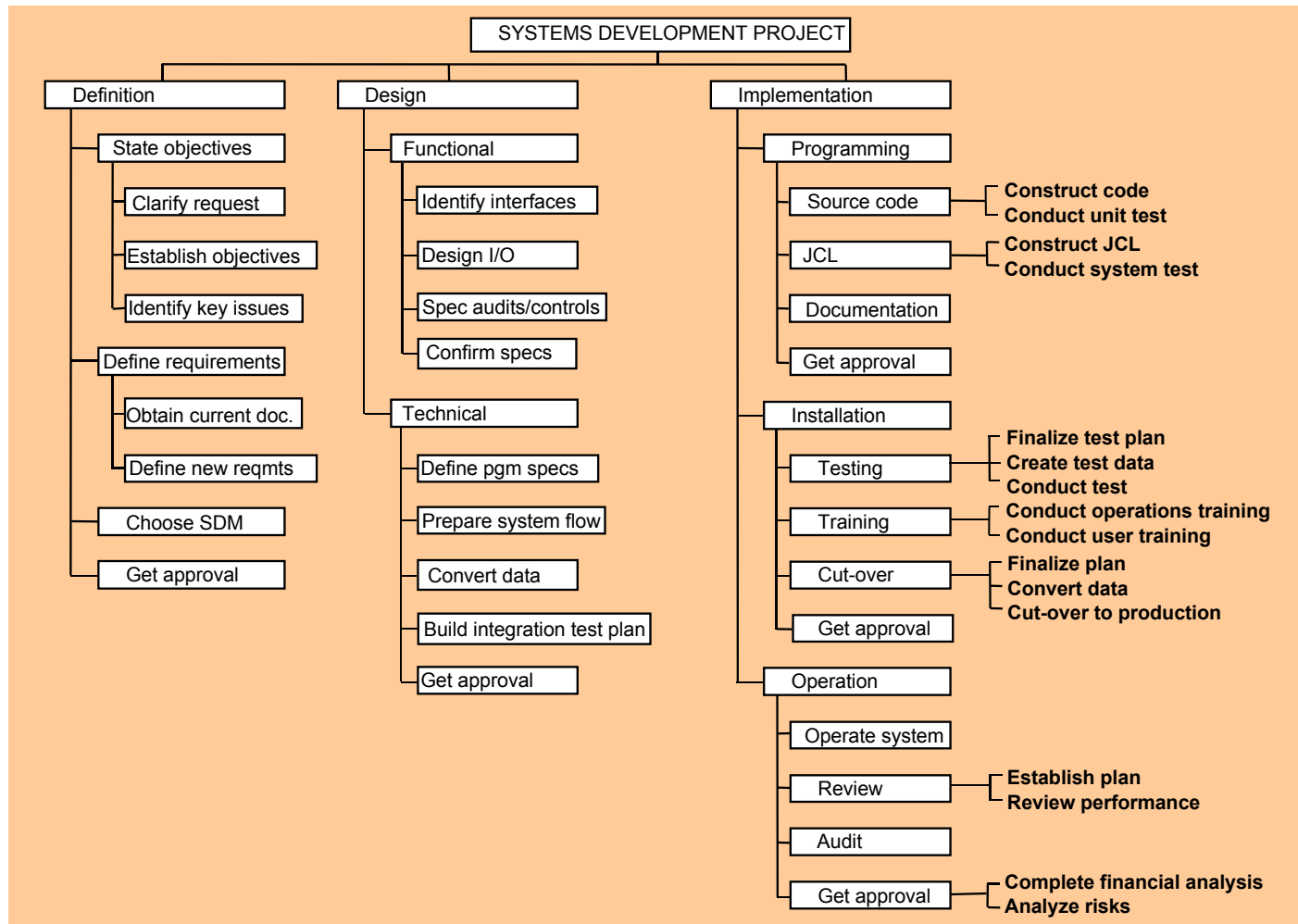
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Case: Indented Outline WBS for a House

1. **SITE PREPARATION**
 - 1.1. **Layout**
 - 1.2. **Grading**
 - 1.3. **Excavation**
2. **FOUNDATION**
 - 2.1. **Erect Forms**
 - 2.2. **Pour Concrete**
 - 2.3. **Remove Forms**
3. **FRAMING**
 - 3.1. **Floor Joists**
 - 3.1.1. **Install first floor joists**
 - 3.1.2. **Install second floor joists**
 - 3.2. **Sub-flooring**
 - 3.2.1. **Install first floor sub-flooring**
 - 3.2.2. **Install second floor sub-flooring**
 - 3.3. **Stud Walls**
 - 3.3.1. **Erect first floor stud walls**
 - 3.3.2. **Erect second floor stud walls**
 - 3.4. **Frame the roof**

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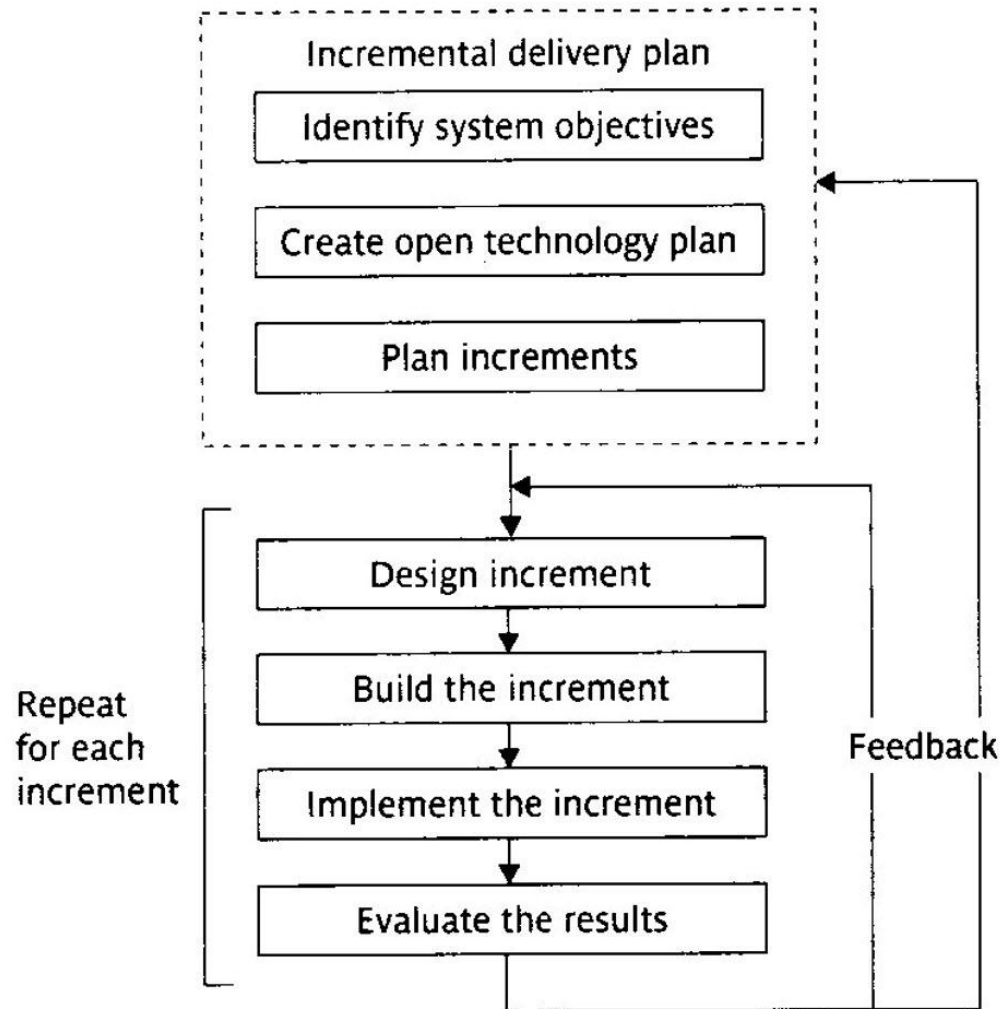
Case: WBS for a Waterfall SDM



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How to generate WBS

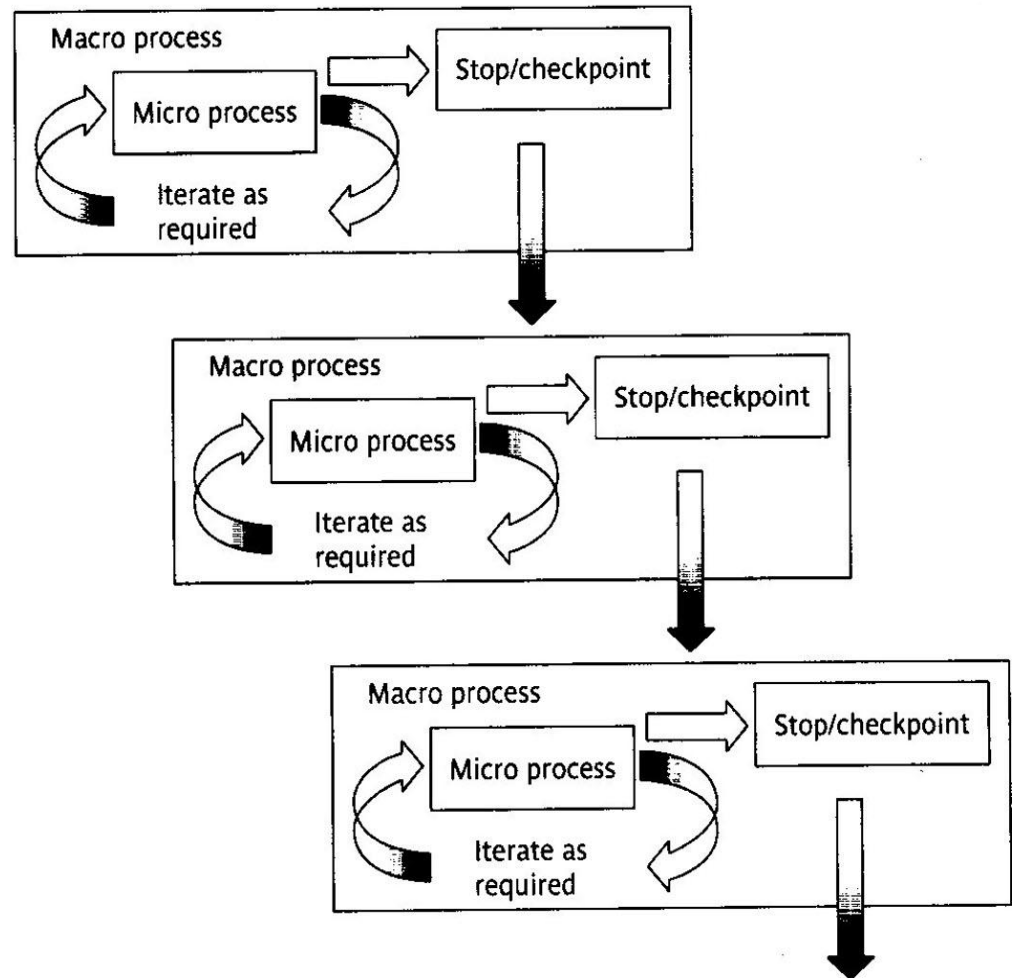
➤ Example



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How to generate WBS

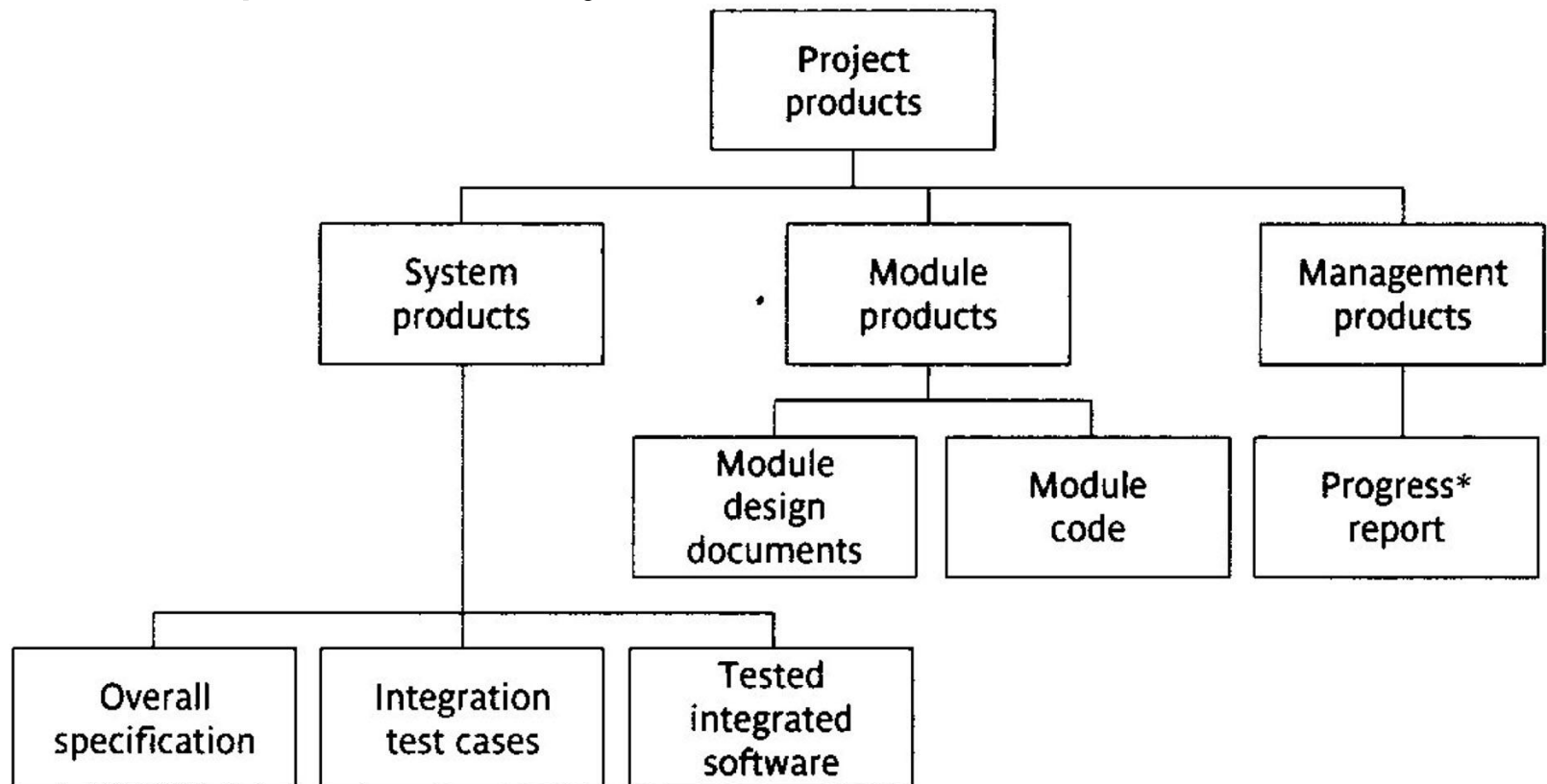
- Example
Macro process



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How to generate WBS

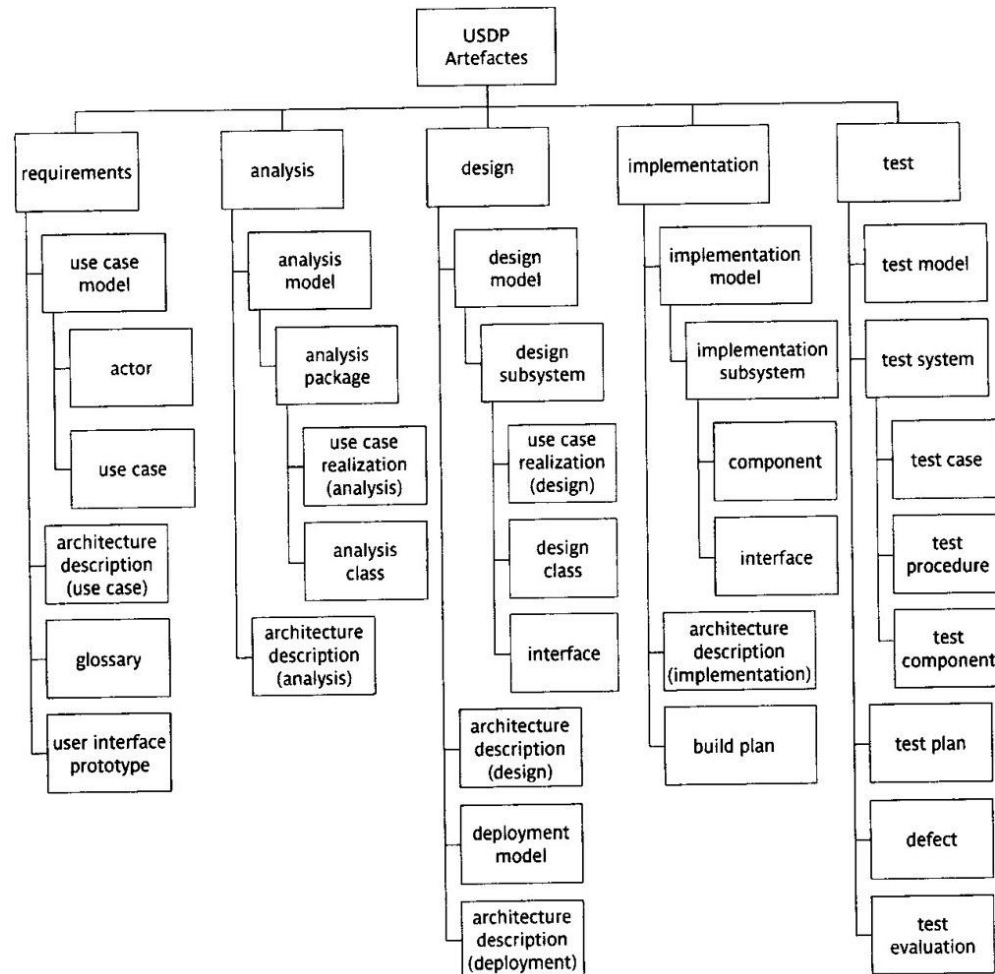
➤ Example: WBS Layout



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How to generate WBS

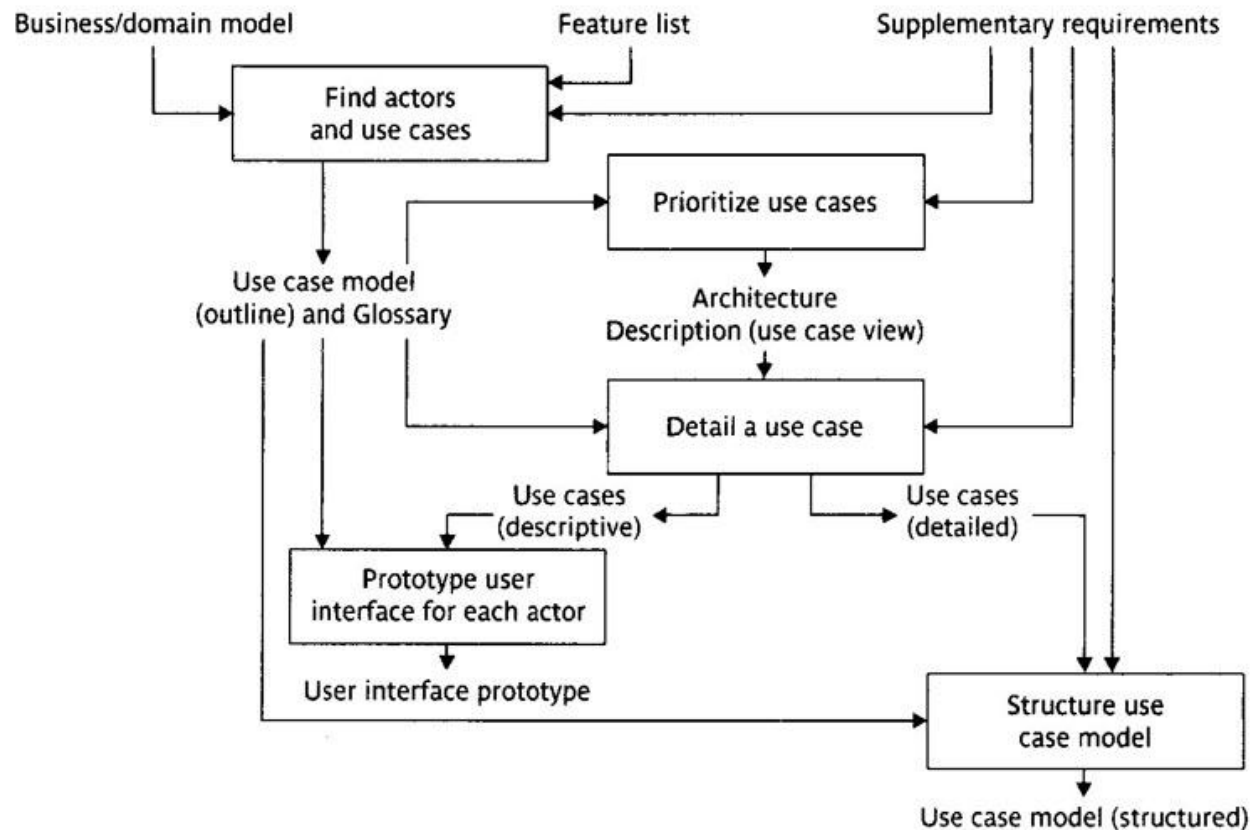
- Example WBS in detail



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How to generate WBS

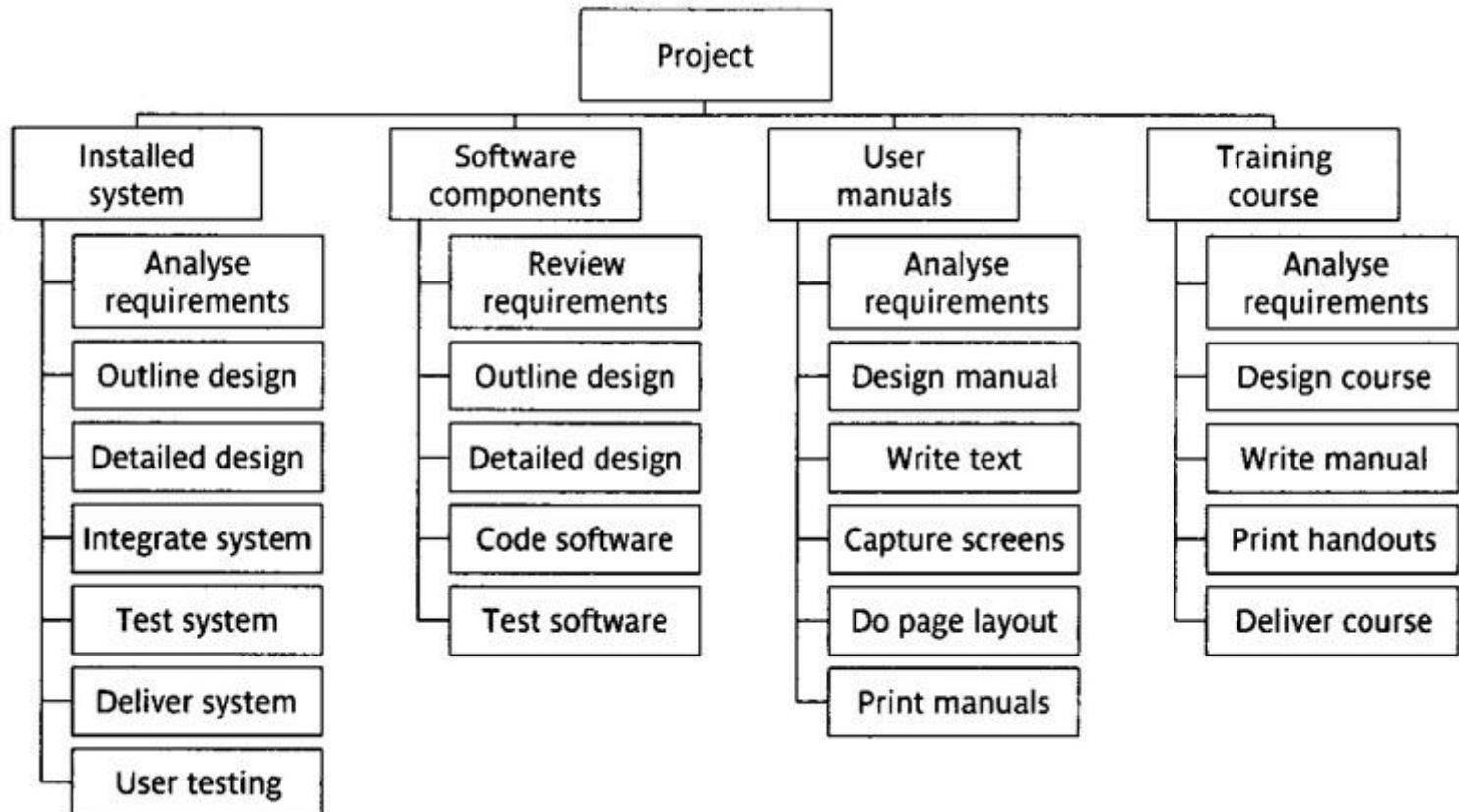
➤ Example: Requirement generation



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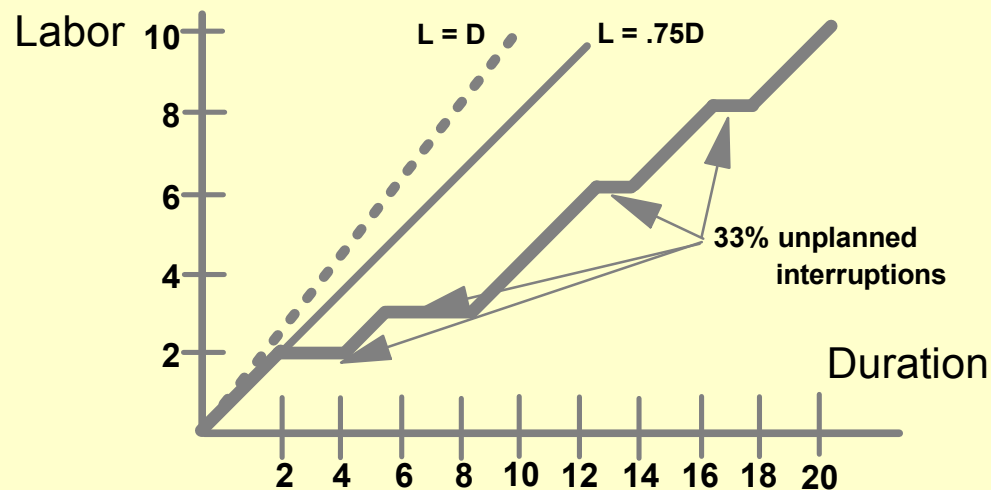
How to generate WBS

- Example: all processes of a project



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Elapsed time (duration) versus work time (labor)



MODEL ASSUMPTIONS

- * Individuals work at 75 percent efficiency rate.
- * Unplanned interruptions account for 33 percent of clock time.

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Should I Estimate Task Duration or Task Labor?

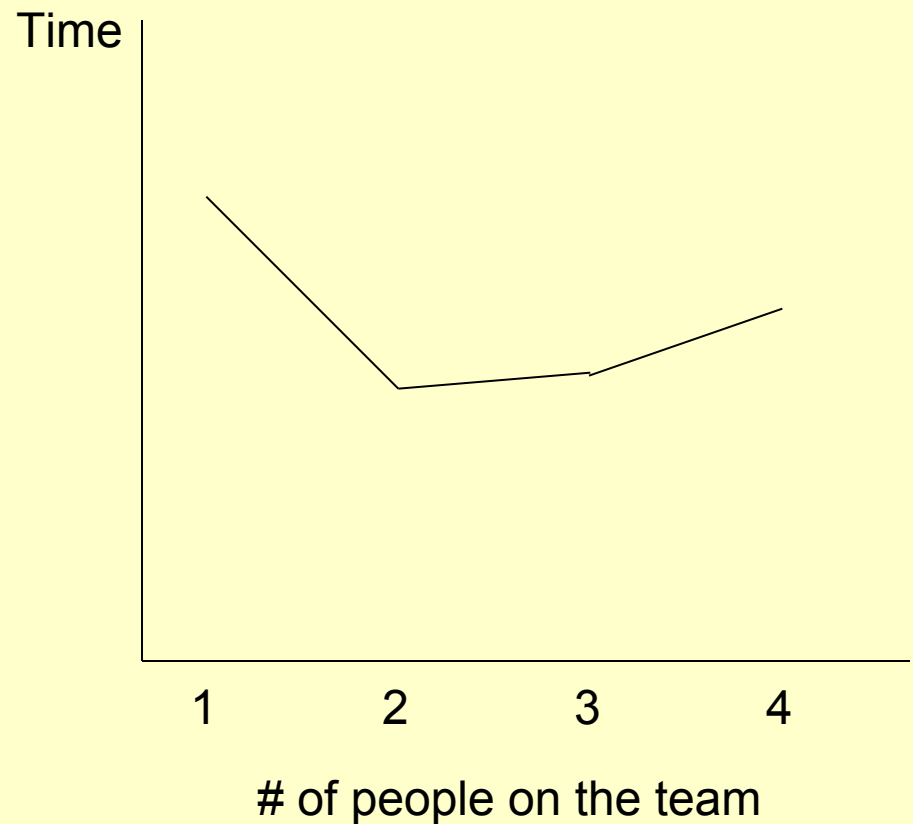
- Estimate duration to build the schedule and determine when the project will be completed
- Estimate labor when you have to contain expenses within a budget



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Case: Resource Loading vs. Task Duration

1. Pick up the chair
2. Carry it to the door
3. Set the chair down
4. Open the door
5. Hold the door open with your foot as you pick up the chair
6. Carry the chair out the door
7. Set the chair in the hallway



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Causes of Variation in Duration

- Varying skill levels
- Unexpected events
- Efficiency of work time
- Mistakes and misunderstandings
- Variation within the capability of the system
- Common cause variation

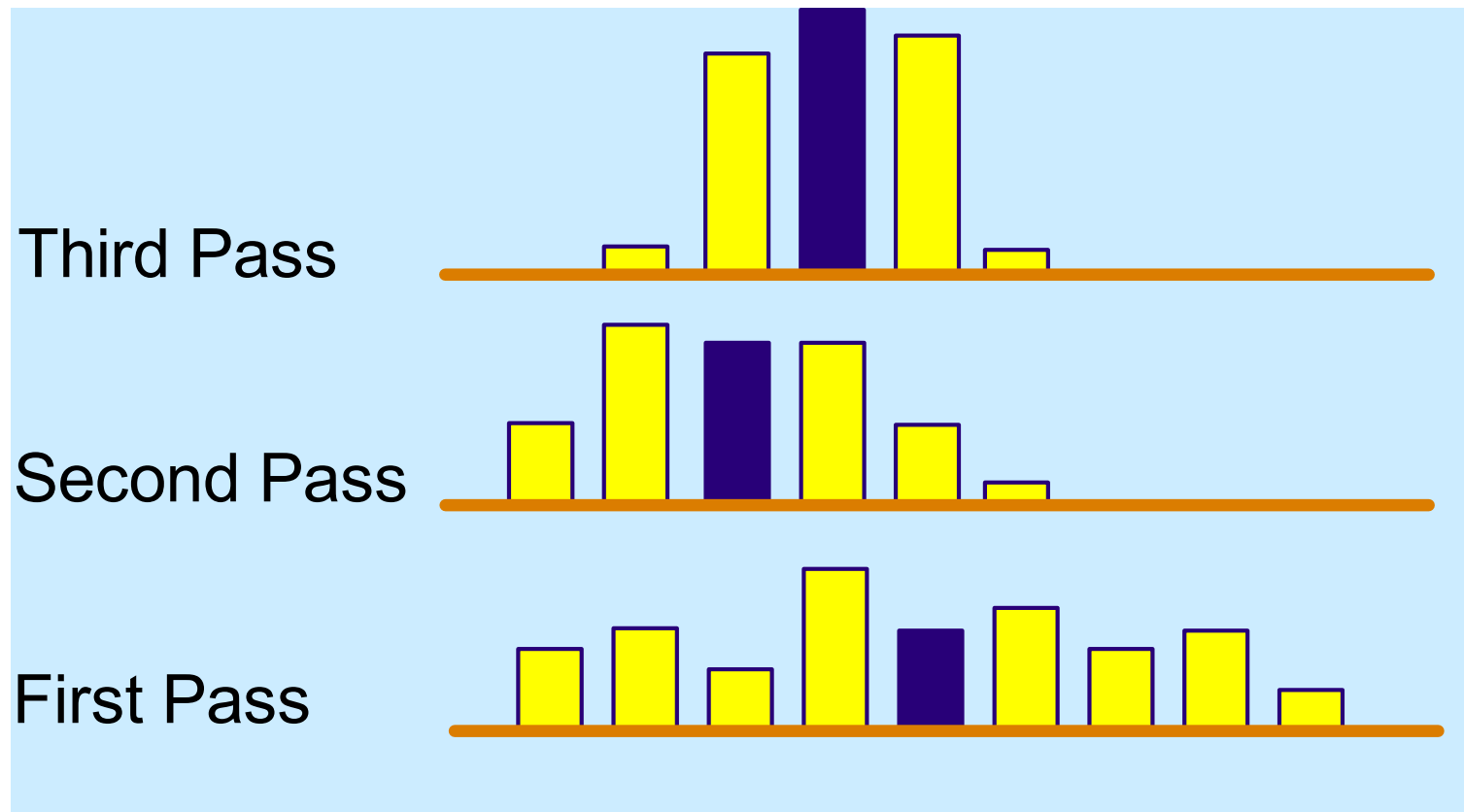
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Six Methods for Estimating Task Duration

- Extrapolating based on similarity to other activities
- Studying historical data
- Seeking expert advice
- Applying the Delphi Technique (next two charts)
- Applying the Three-point technique (next two charts)
- Applying the Wide-band Delphi technique

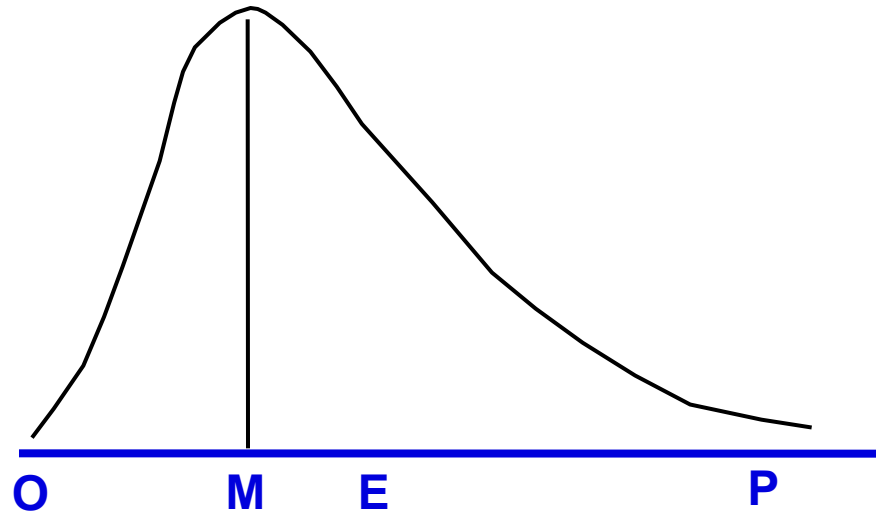
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The Delphi Technique



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The Three-Point Method



O: Optimistic

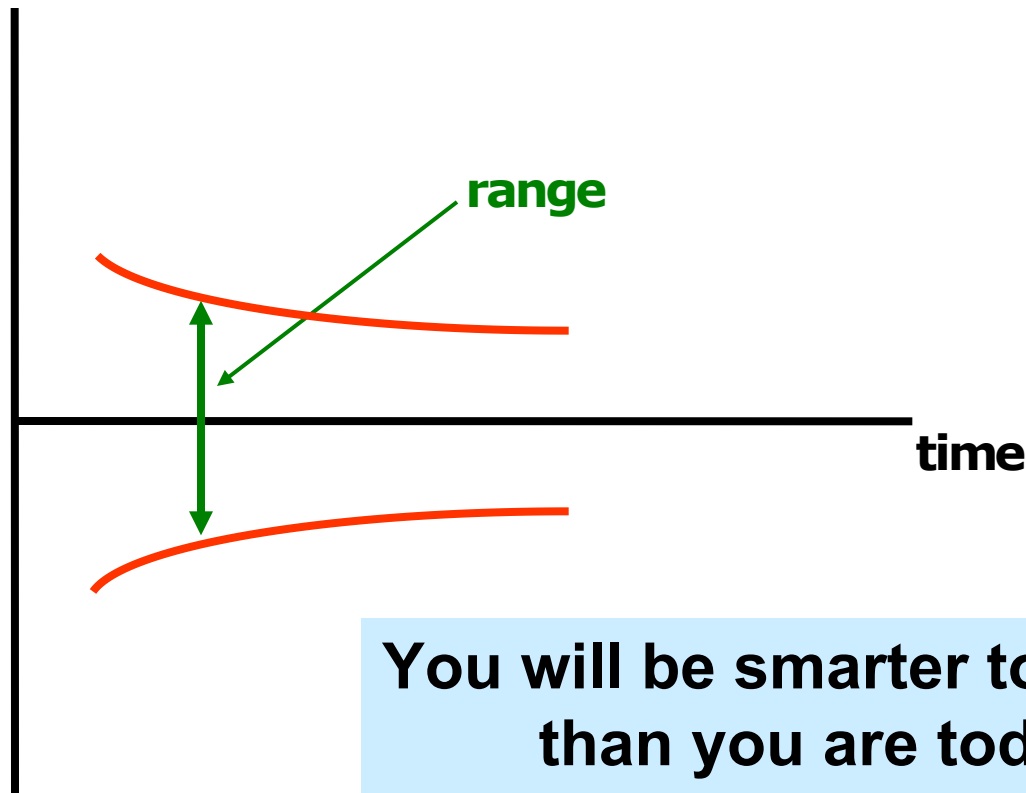
P: Pessimistic

M: Most Likely

$$E = \frac{O + 4M + P}{6}$$

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Estimates Have a Life Cycle Too



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Estimating Resource Requirements

- People
- Facilities
- Equipment
- Money
- Materials

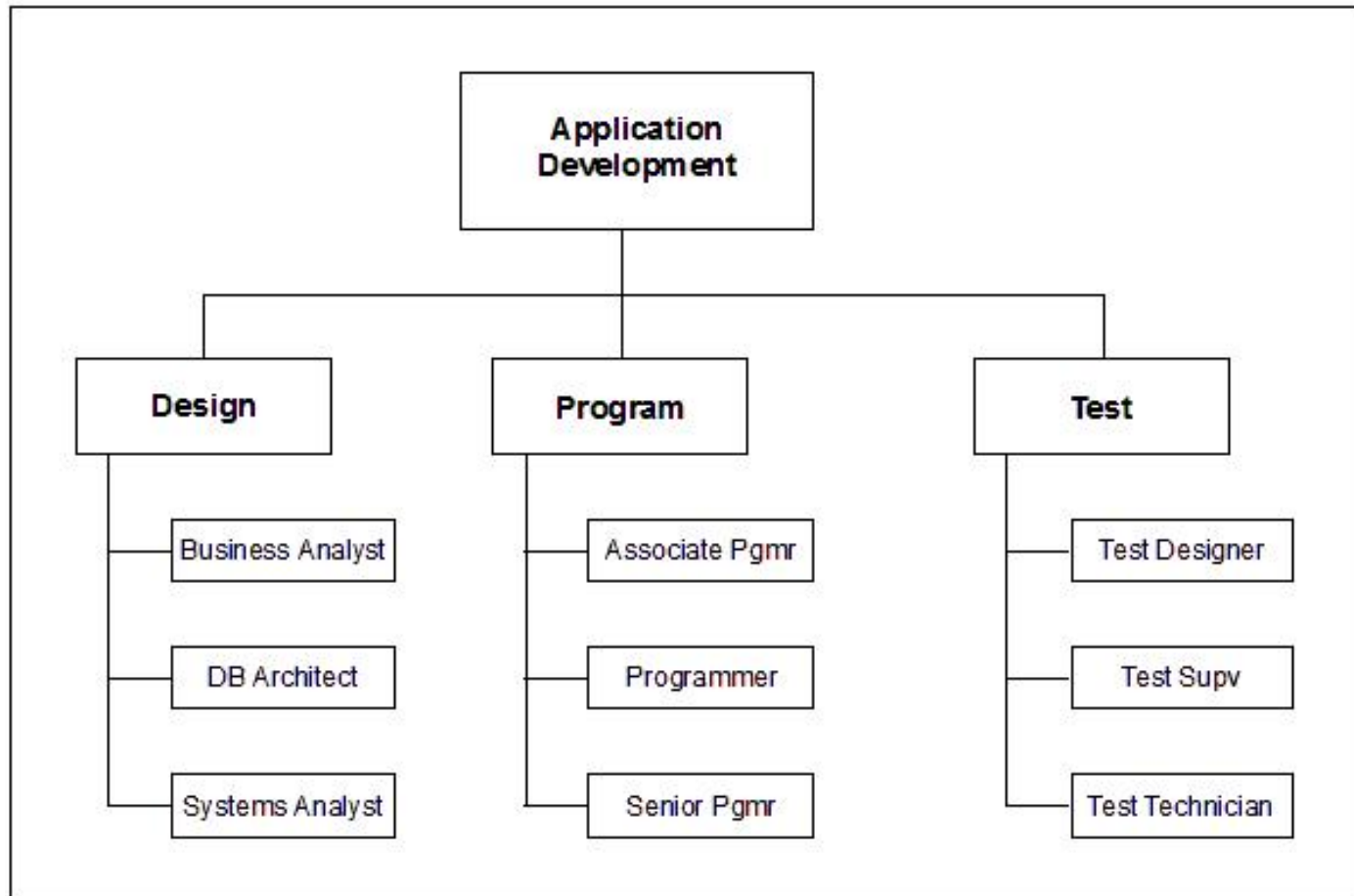
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Assigning Staff to Activities



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Resource Organizational Structure



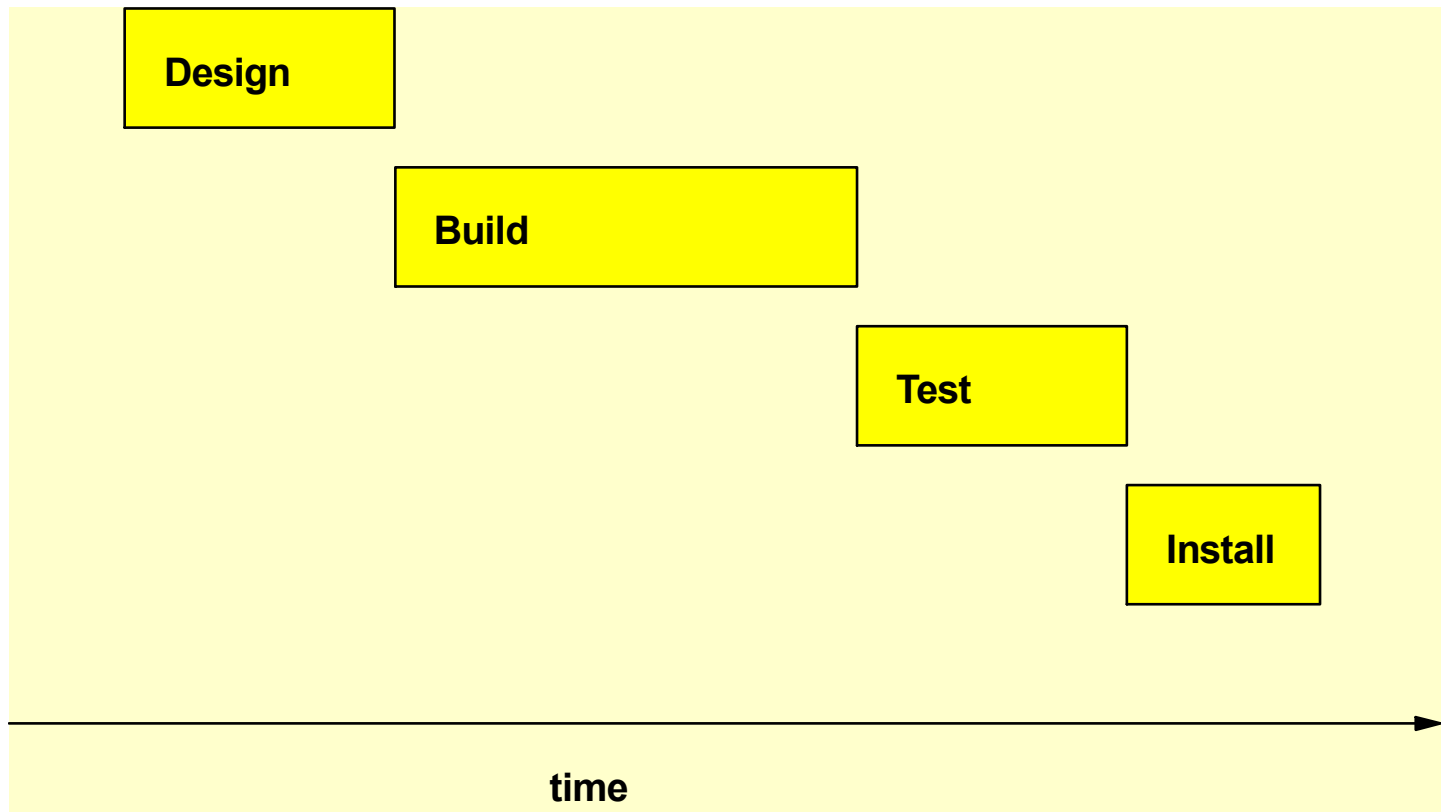
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Estimating Cost

- Order of magnitude estimate
- Budget estimate
- Definitive estimate
- Cost Budgeting
- Cost Control
 - Weekly reports
 - Baseline versus actual

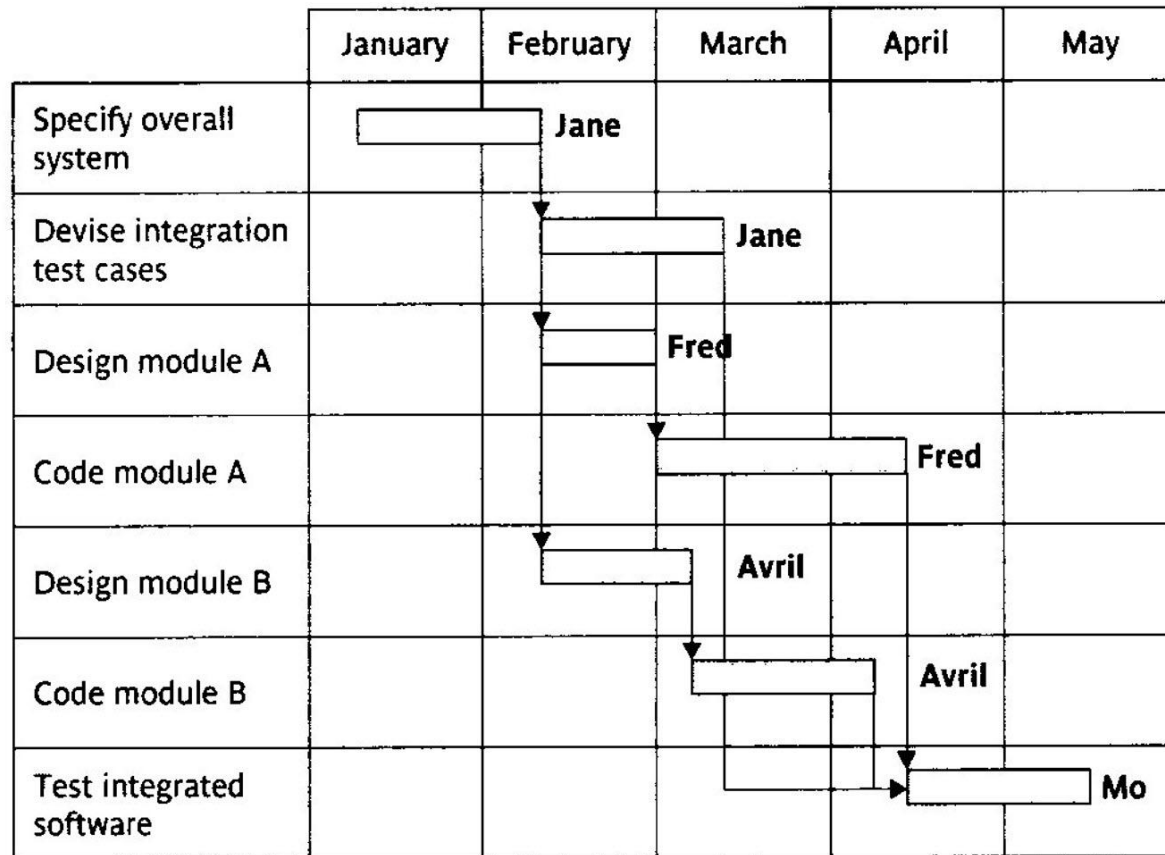
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Network-based Scheduling – Gantt Charts



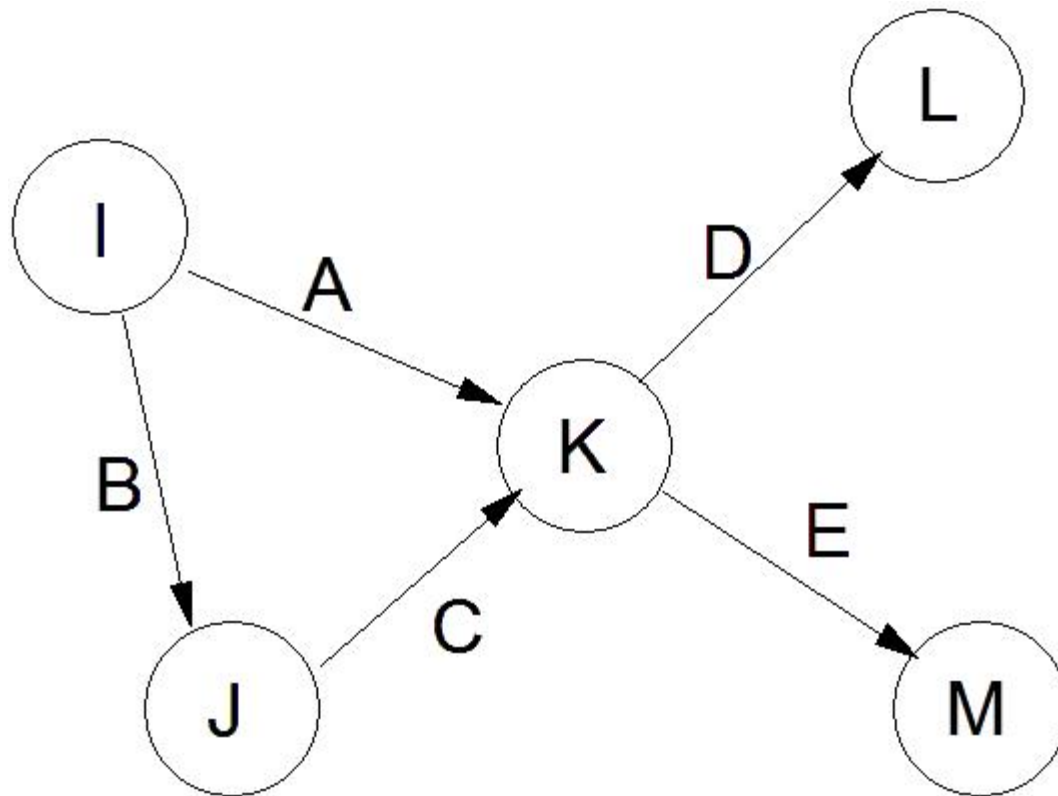
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Network-based Scheduling – Gantt Charts



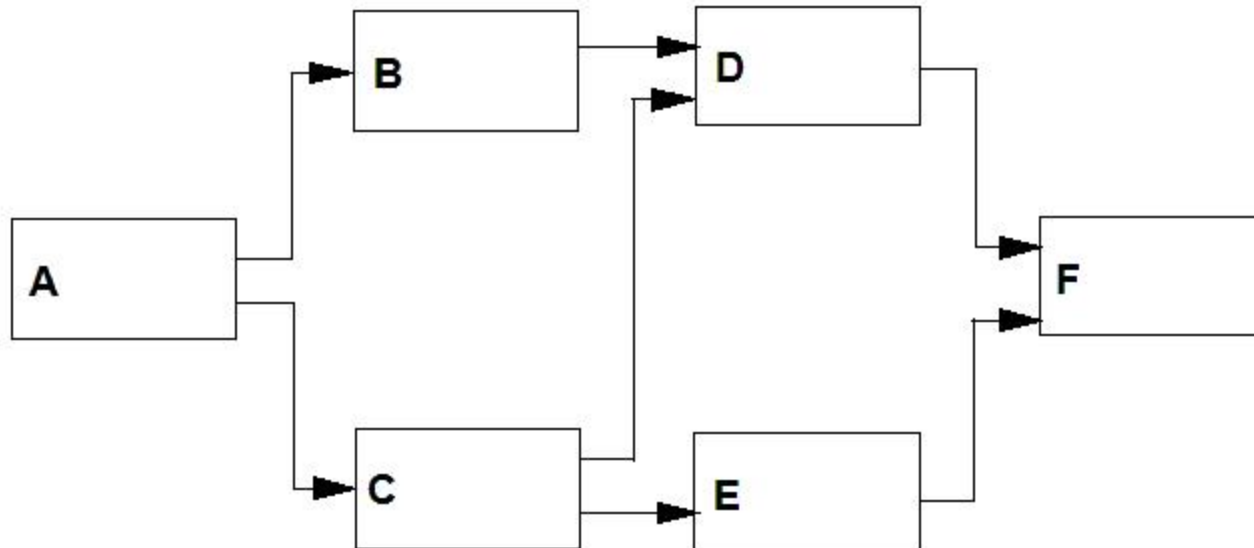
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Network-based Scheduling – Task on the Arrow



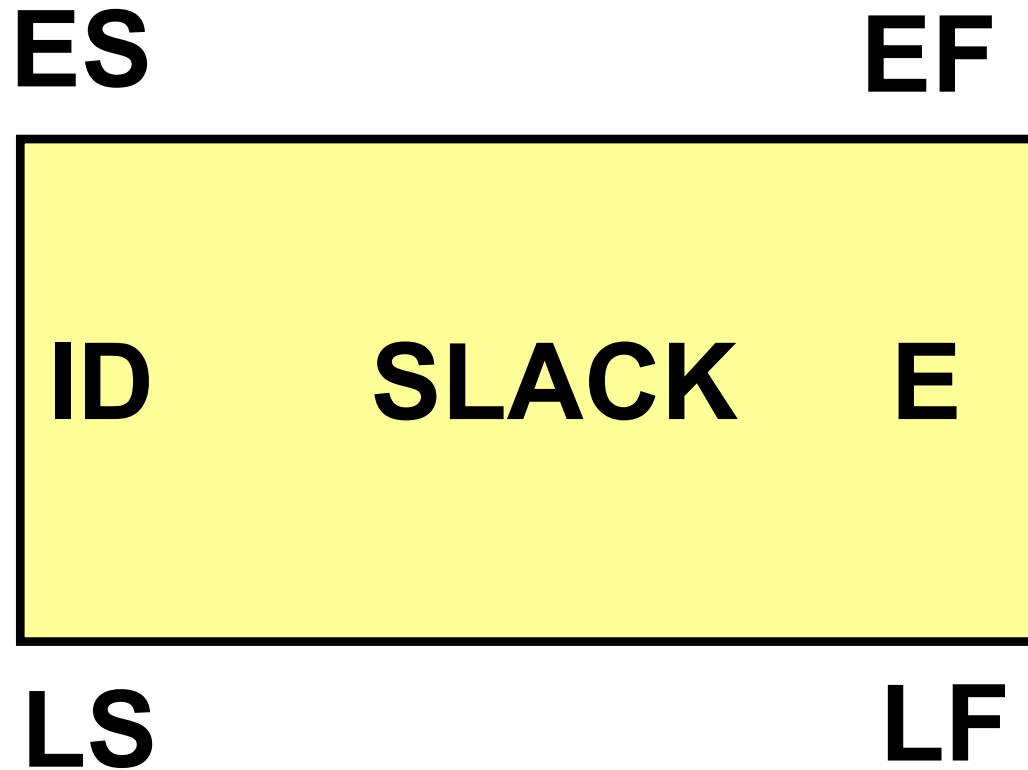
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Network-based Scheduling – Task on the node format



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Task Node



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Case of critical path calculation

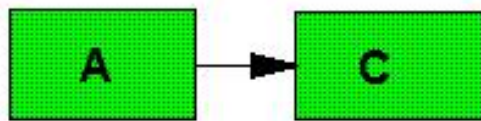
➤ Node Definition

ES	Duration	EF
ID	Activity Description	
LS	Slack	LF

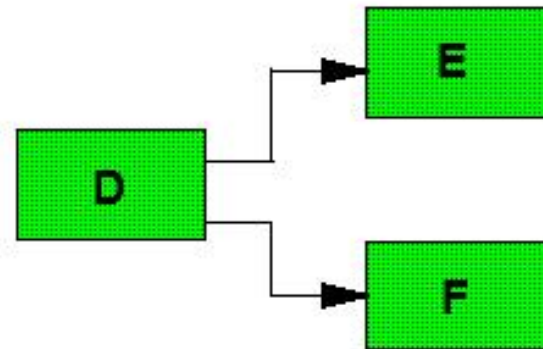
最早开始	周期	最早完成
活动标签，活动描述		
最迟开始	缓冲期	最迟完成

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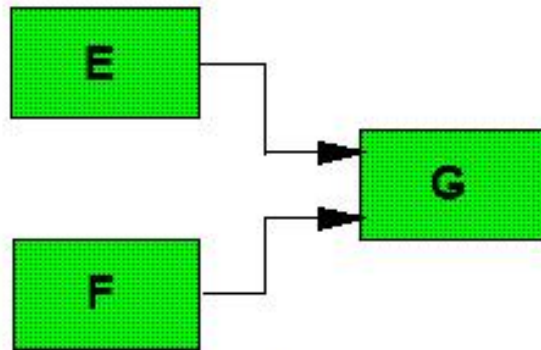
Diagramming Conventions



(a)



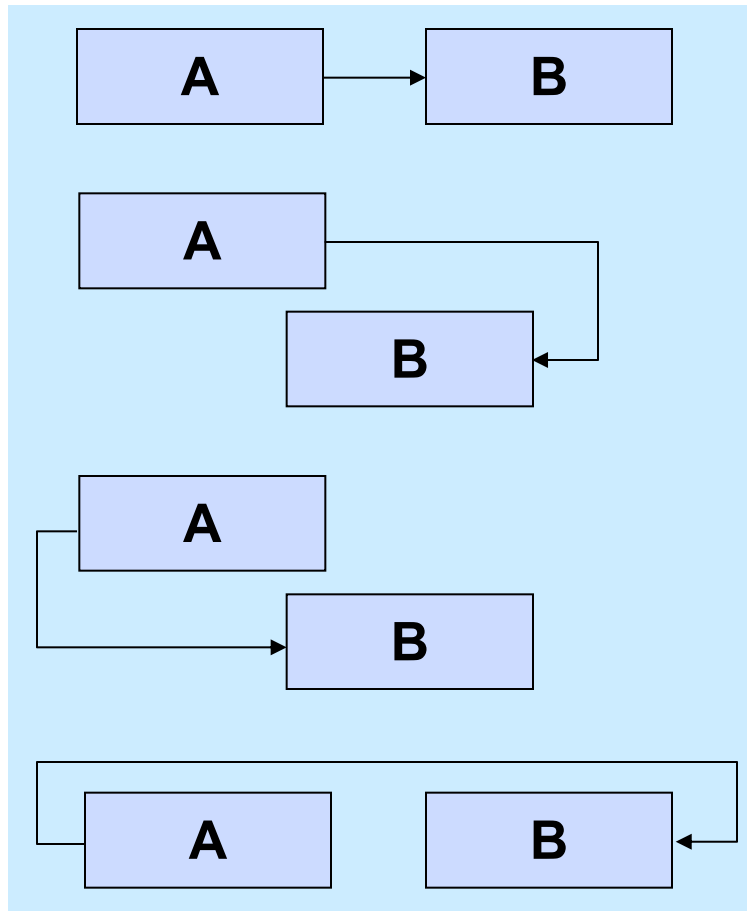
(c)



(b)

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Dependency Relationships



FS: When A finishes, B may start

FF: When A finishes, B may finish

SS: When A starts, B may start

SF: When A starts, B may finish

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Dependency Constraints

- Technical constraints
 - Discretionary constraints
 - Best practice constraints
 - Logical constraints
 - Unique requirements constraints
- Management constraints
- Inter-project constraints
- Date constraints
 - No earlier than
 - No later than
 - On this date
- Lag variables

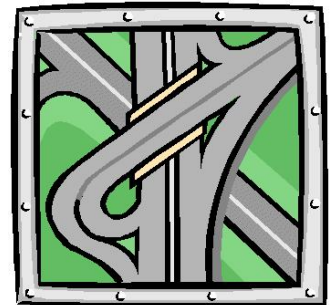


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Paths Through the Network

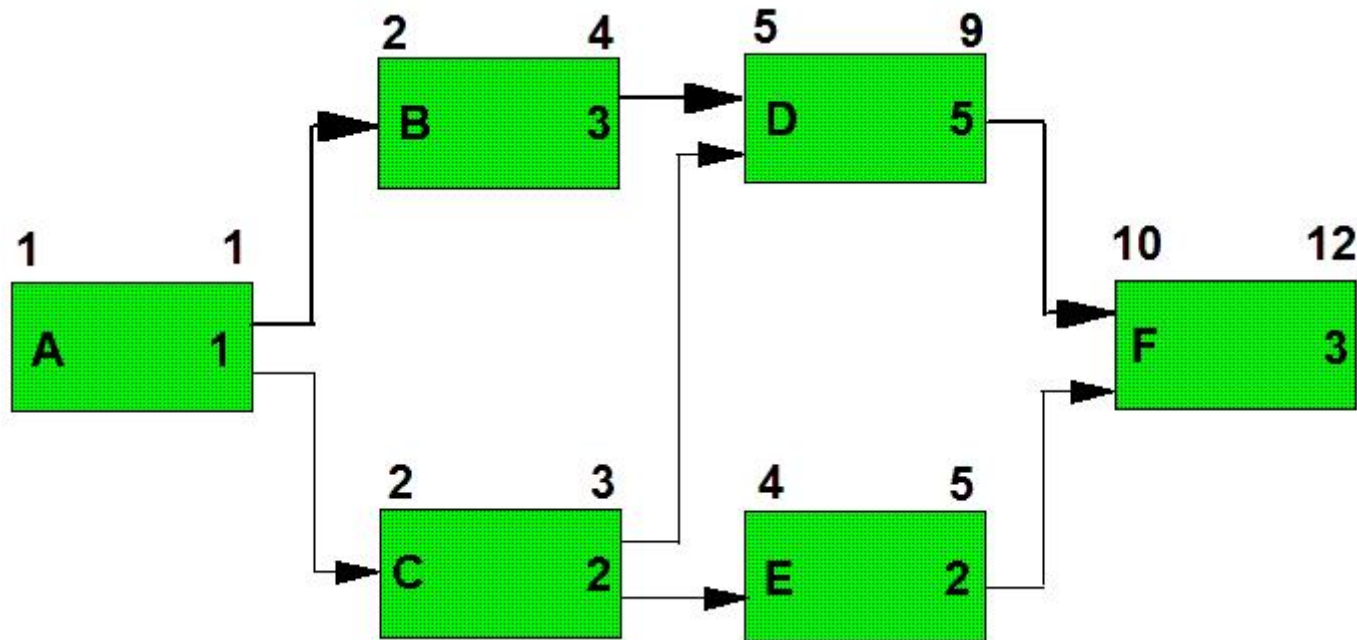
- The Forward Pass – Early Schedule
 - Left to right (start to finish)
 - Determines Early Start and Early Finish
 - $ES + \text{duration} - 1 = EF$

- Backward Pass – Late Schedule
 - Right to left (finish to start)
 - Determines Late Start and Late Finish
 - $LF - \text{duration} + 1 = LS$



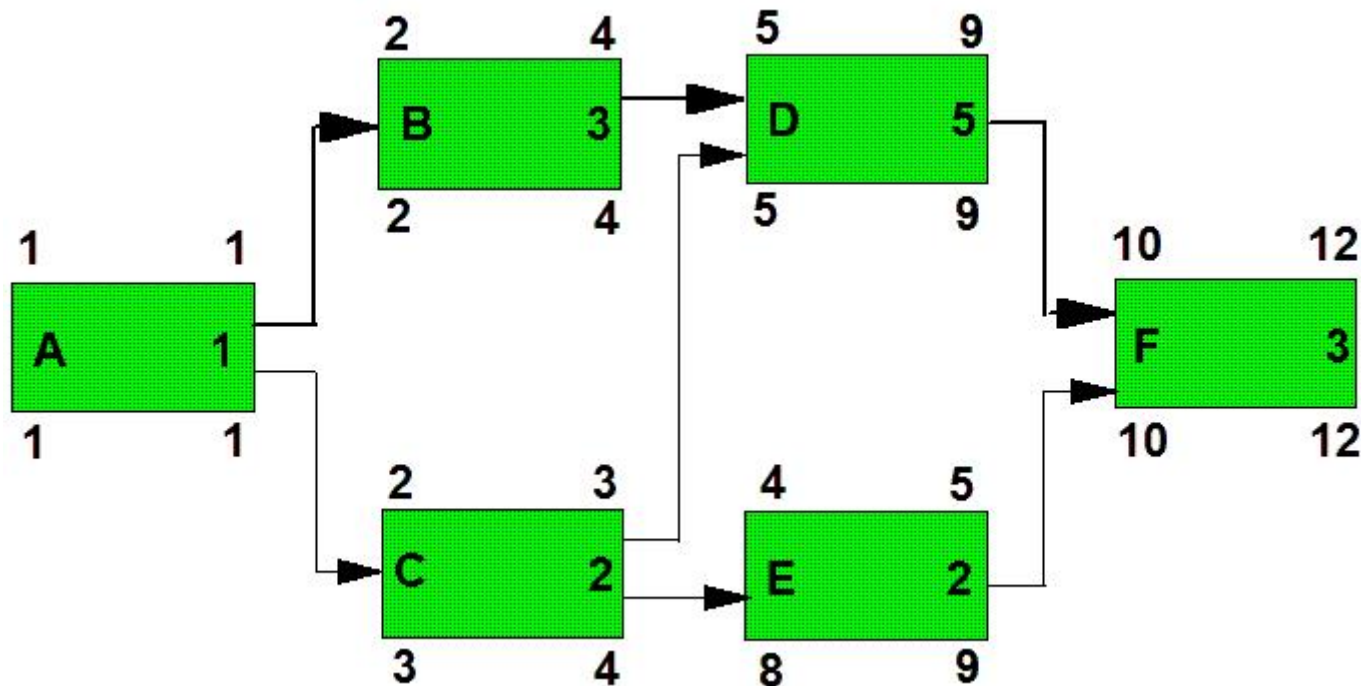
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Forward Pass Calculations – Early Schedule



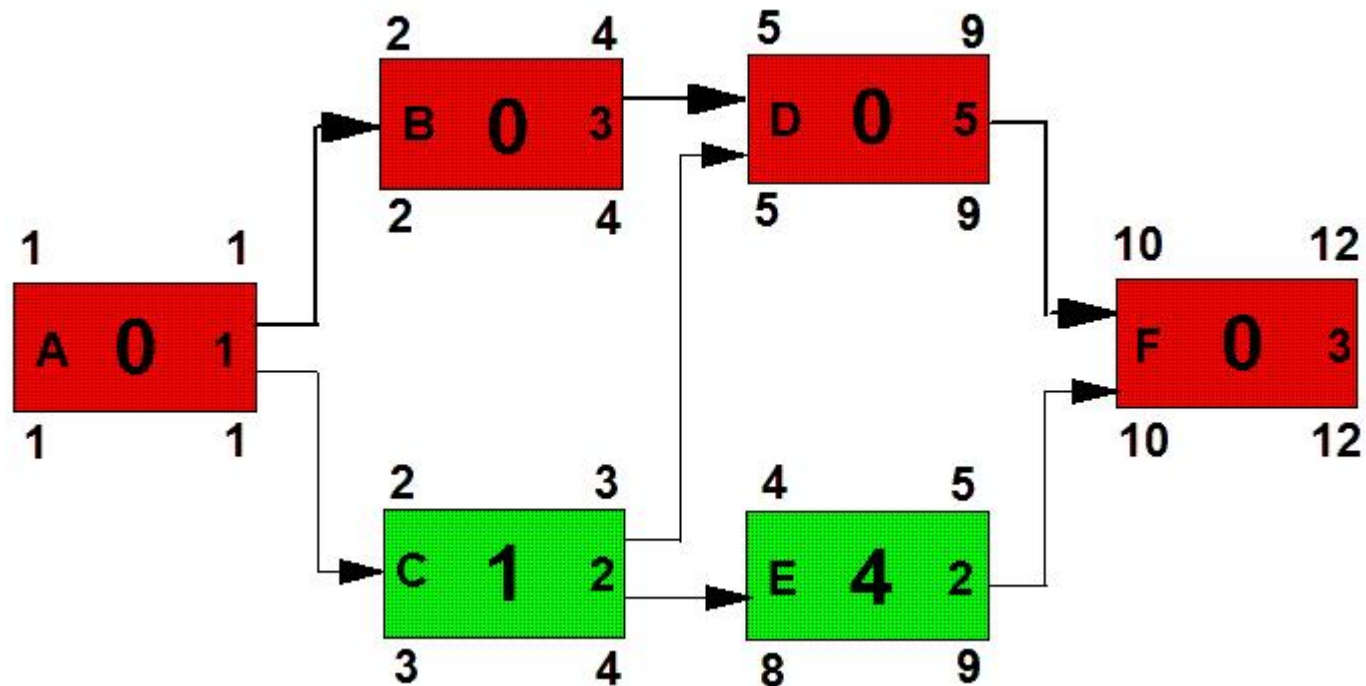
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Backward Pass Calculations – Late Schedule



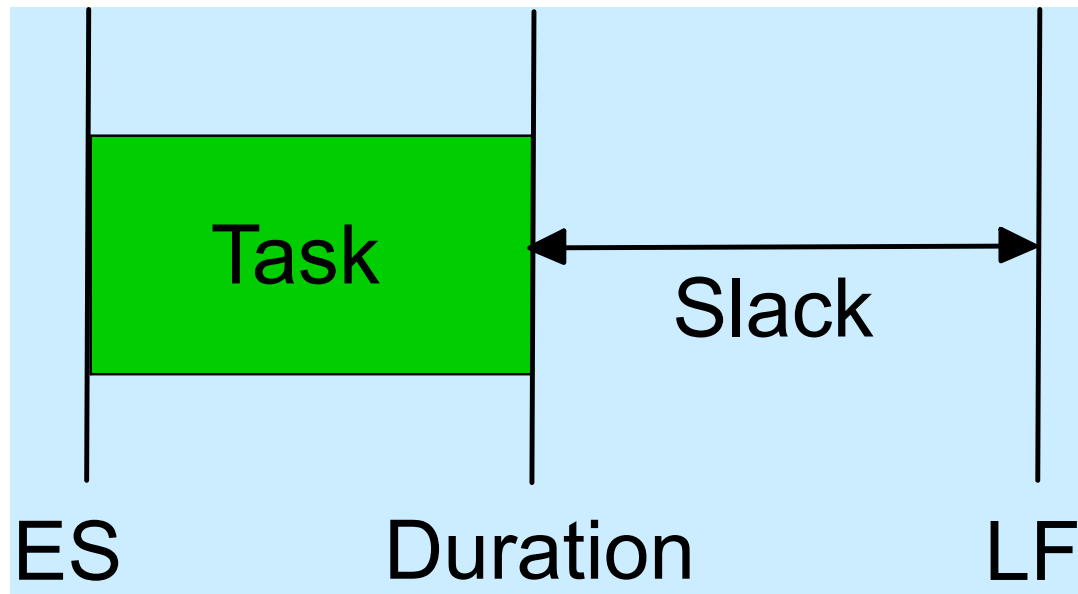
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Critical Path



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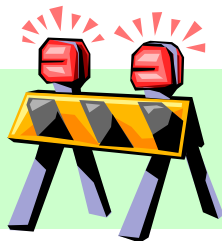
Early Start to Late Finish Window



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Slack Time

- Total Slack
 - Time that a task can be delayed without impacting the early schedule of the project.
- Free Slack
 - Time a task can be delayed without impacting the early schedule of its successor tasks.



**Do Not Plan to Use Slack
to Bail Out(挽救) the Project!**

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Building the Initial Dependency Diagram

1. Enter task name and duration into software tool
2. Print PERT Diagram
3. Cut out each task node and affix to a Sticky Note
4. Write task ID # on Post-It Note
5. Place Sticky Notes on right side of whiteboard
6. Position start node on left side of whiteboard
7. Move all tasks with no predecessor to left side and connect to start node
8. Move all tasks on the right side with predecessors on the left side to the left side and connect with single line
9. Continue until all task nodes have been moved to left side

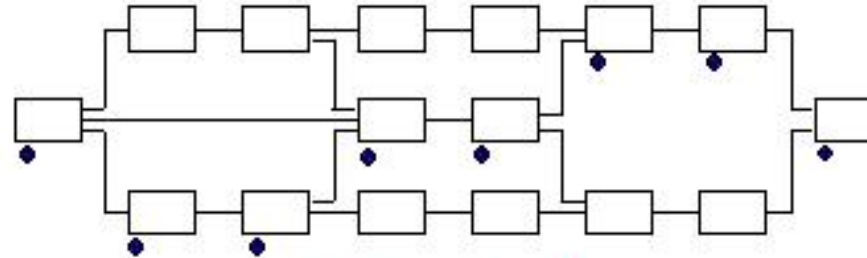
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Schedule Compression Techniques

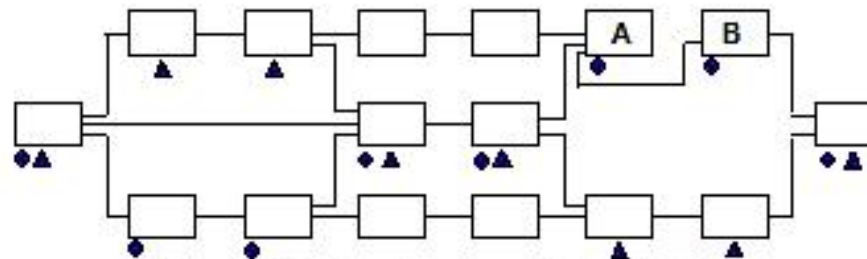
- Replace FS with SS dependencies
- Replace a team member with a more skilled person
- Add resources
 - From non-critical path tasks to critical path tasks
 - From other projects
 - To where
 - critical path tasks
 - high-risk tasks
 - tasks with large duration variances

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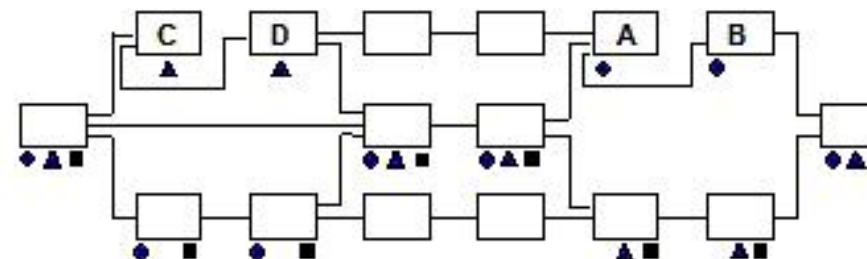
Schedule Compression Iterations



Original critical path



Critical path after changing AB from FS to SS



Critical path after changing CD from FS to SS

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Case: critical path calculation

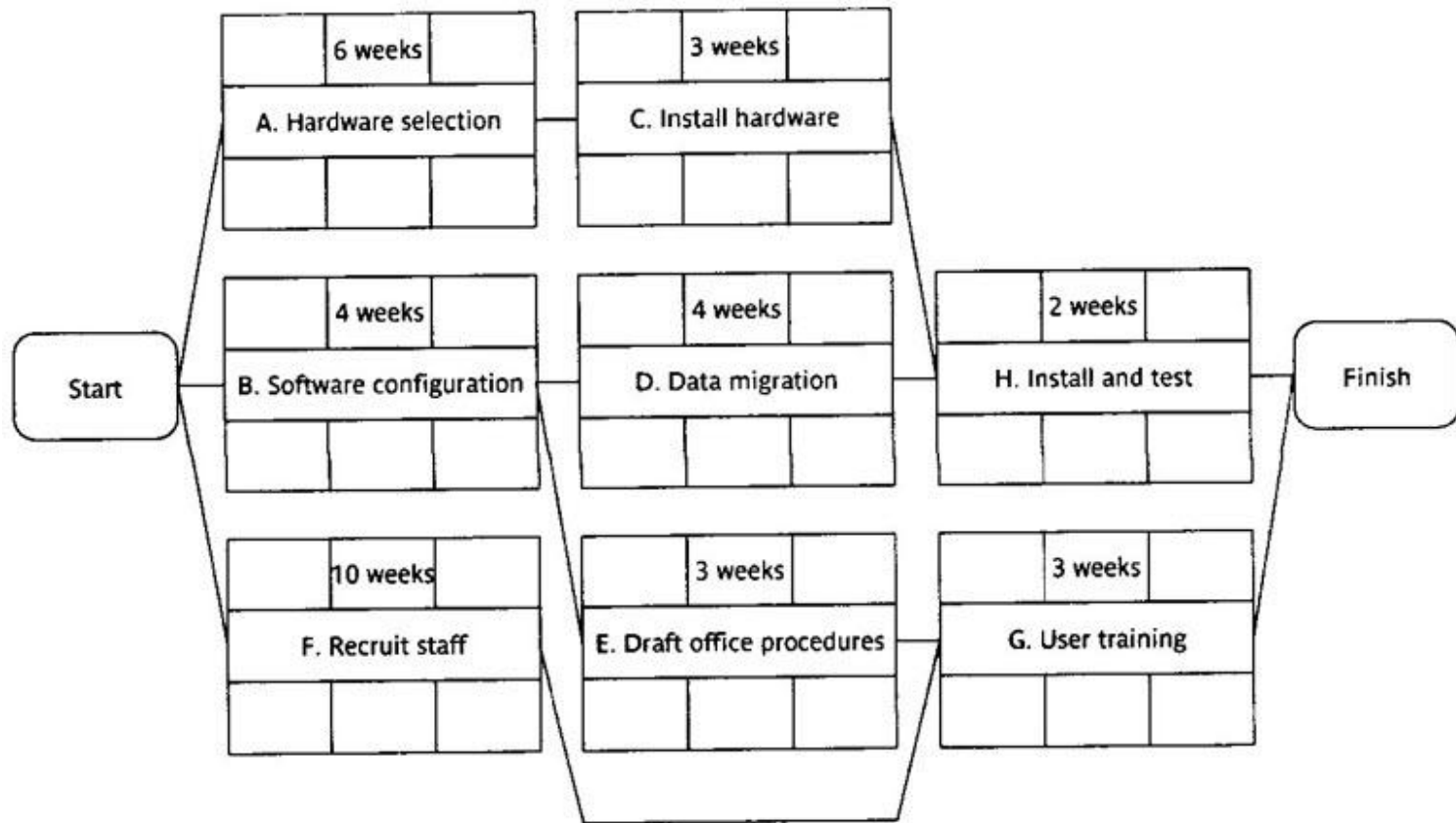
➤ Relationship of activities

Activity	Duration (weeks)	Precedents
A Hardware selection	6	
B System configuration	4	
C Install hardware	3	A
D Data migration	4	B
E Draft office procedures	3	B
F Recruit staff	10	
G User training	3	E, F
H Install and test system	2	C, D

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Case: critical path calculation

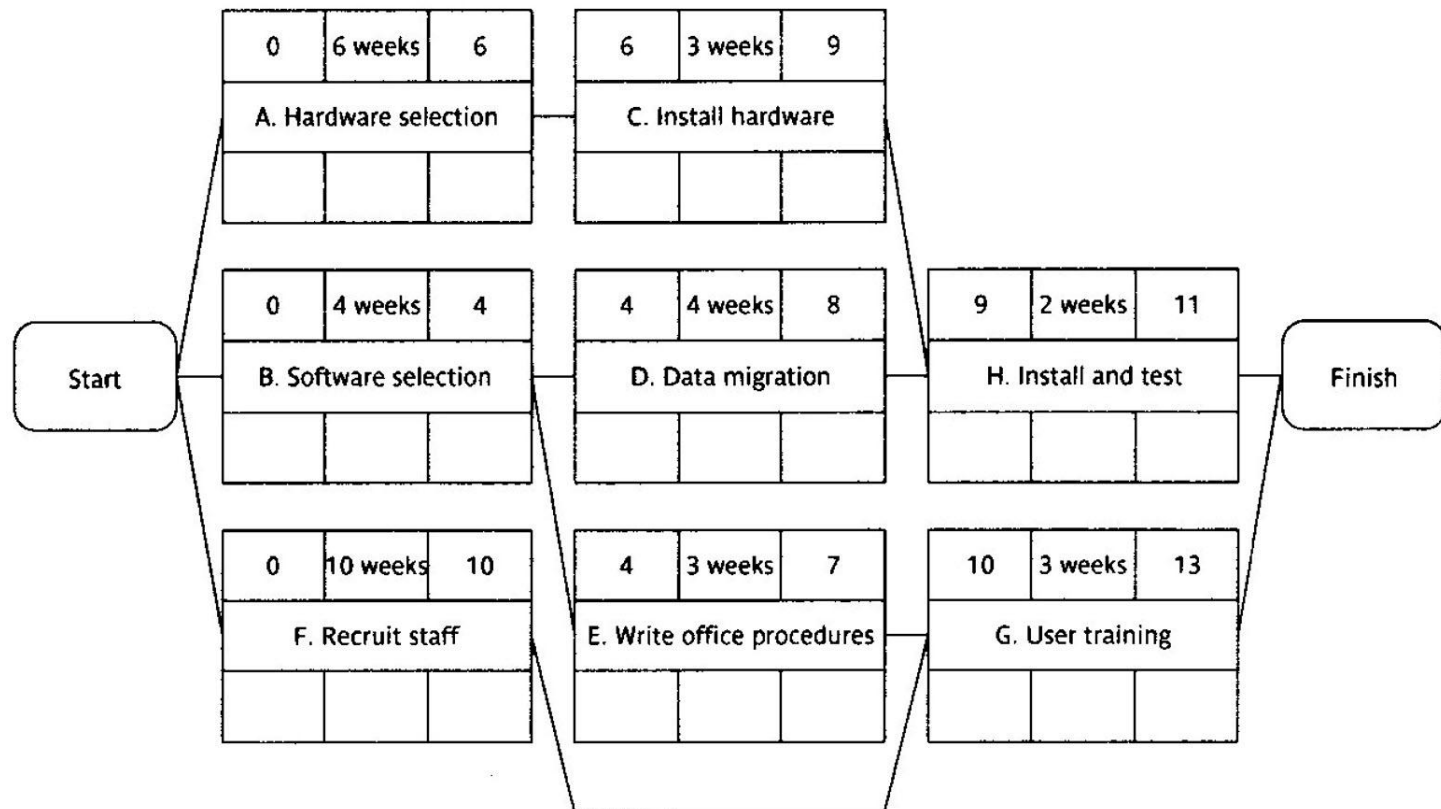
➤ Network of activities



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Case: critical path calculation

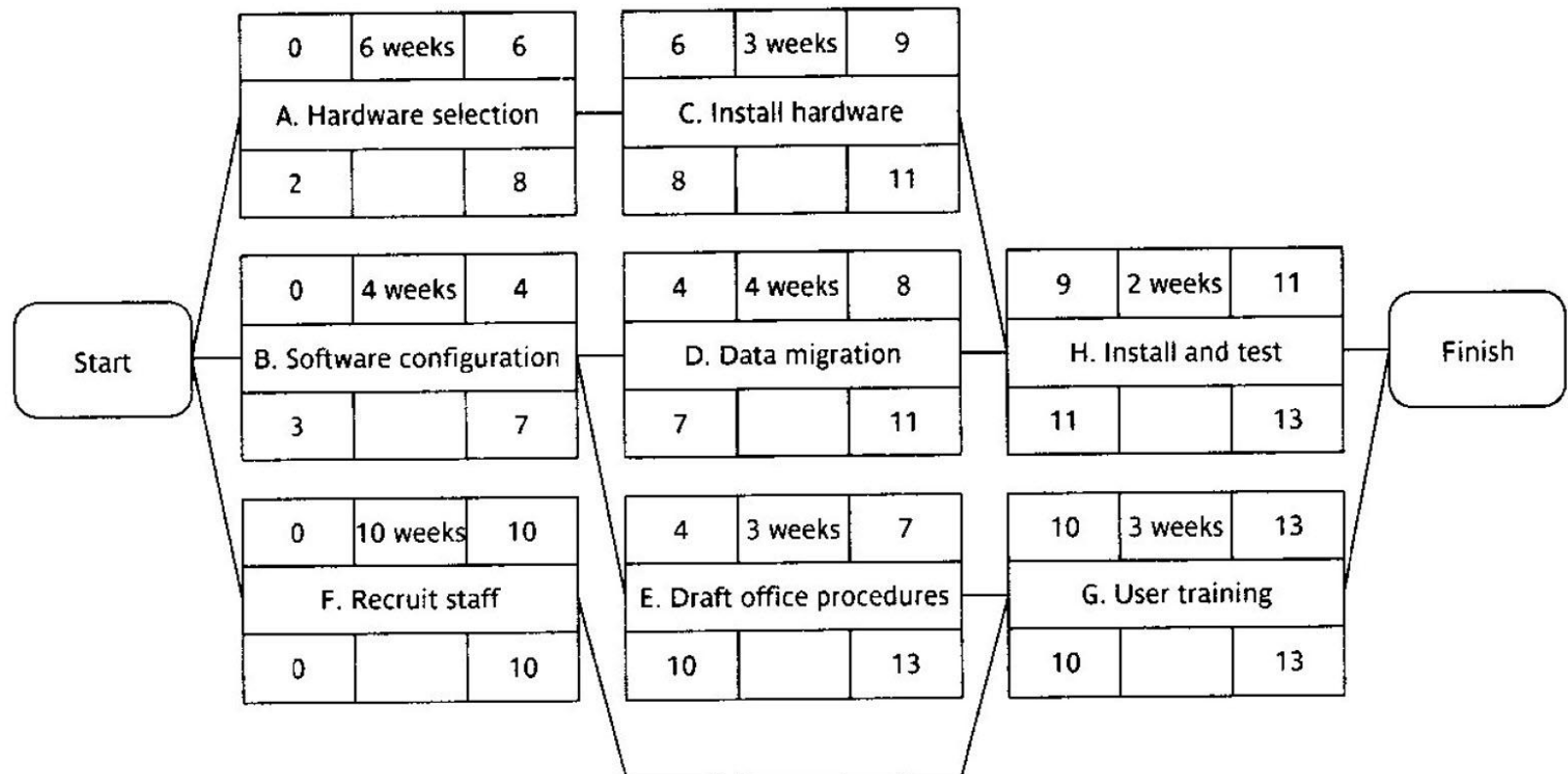
➤ Forward calculation



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Case: critical path calculation

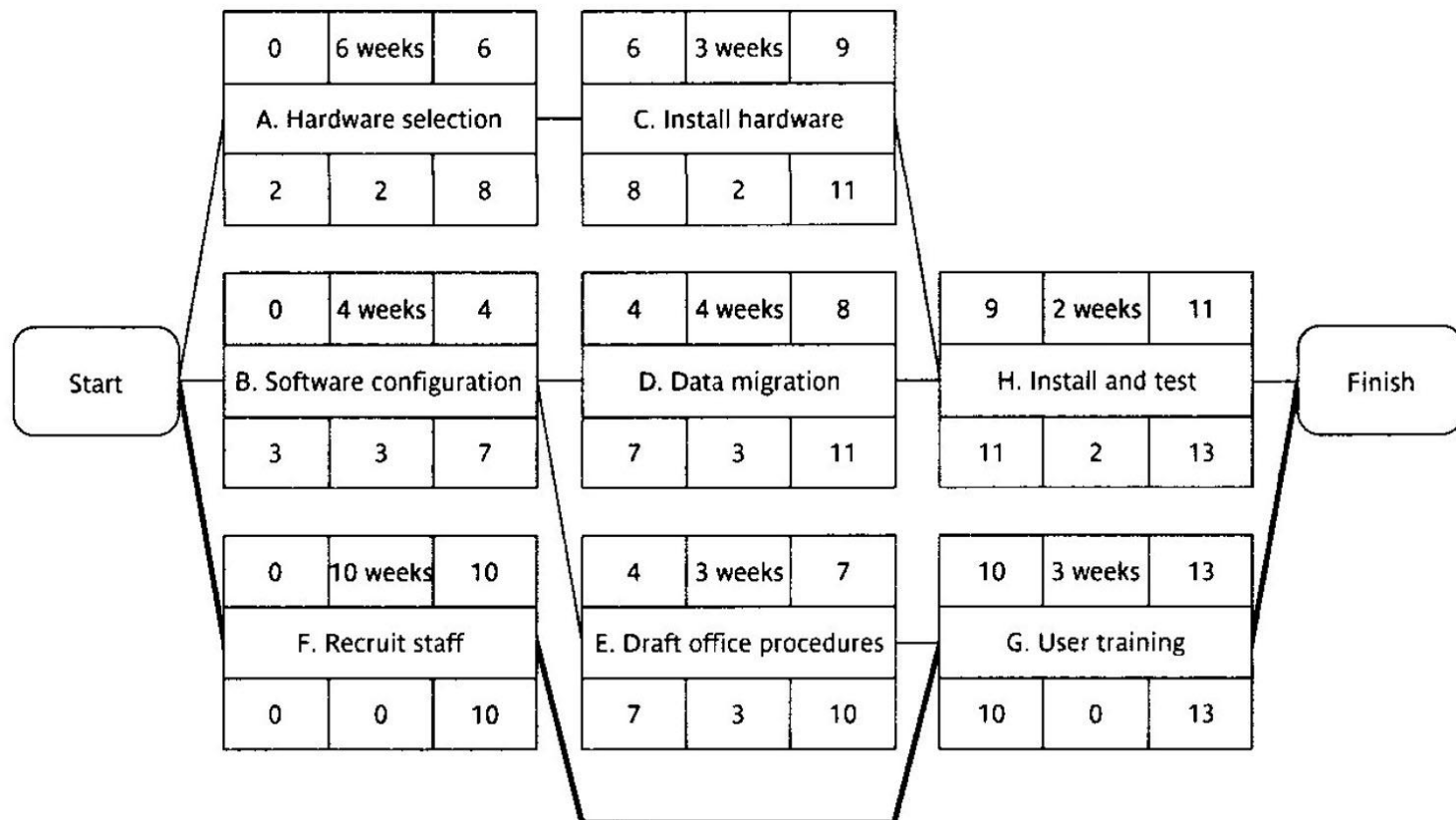
➤ Backward calculation



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Case: critical path calculation

➤ Critical path



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Contents of the Project Proposal

- Executive Summary
- Background
- Objective
- Overview of the approach to be taken
- Detailed statement of work
- Time and cost summary
- Appendices

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Gaining Approval to Launch the Project

- The cost/benefit is not in your favor.
- The risks of failure are too high.
- The total project cost exceeds available funding.
- There are other projects competing for the same resources.

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Summary of Lesson 4

- Using tools, templates, and processes to plan a TPM project
- The importance of planning
- Using application software packages to plan projects
- Planning & conducting the Joint Project Planning Session (JPPS)
- Building the Work Breakdown Structure (WBS)
- Estimating task duration
- Estimating resource requirements
- Estimating cost
- Constructing the project network diagram
- Writing an effective project proposal
- Gaining approval to launch the project

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Assignment I of Lesson Four

Create a precedence activity network using the following details:

Activity	Depends on	Duration (days)
A		5
B	A	7
C	B	6
D	A	5
E	D	10
F	B	15
G	B	8
H	G	8
I	C	4
J	G	4
K	E, F	5
L	I, H	3

Calculate the earliest, latest start and end dates and the float associated with each activity in the network you have created above. From this identify the critical path.

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Assignment **II** of Lesson **Four**

- Each team should design and write the **Project Charter** based on the characteristics of your course project.
- The contents of **Project Charter** should be written according to the template on the textbook.
- The team leader should submit **Project Charter** of your team on canvas platform before **29th March, 2026**.